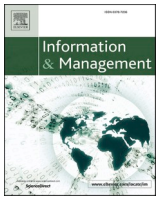
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Advancing an LDA-GMM-CorEx topic model with prior domain knowledge in information systems research

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ABSTRACT

Embedding topic models with domain knowledge is deemed to be effective in bolstering the models' interpretability. Nevertheless, contemporary topic modeling techniques introduced in past studies lack consideration for circumstances in which prior domain knowledge either does not exist or becomes obsolete quickly. Combining the latent Dirichlet allocation (LDA) with the Gaussian mixture model (GMM) and the anchor correlation explanation (CorEx) topic model, we advanced a novel LDA-GMM-CorEx topic modeling approach to enhance the domain knowledge model's adaptability and improve the interpretability of topic modeling. We further verified the effectiveness of our proposed topic modeling approach on two separate datasets from different domains, thereby attesting to its general applicability.

1. Introduction

With the burgeoning amount of user-generated data, like customer reviews [1,2] and tweets [3], being produced constantly, scholars have turned to topic modeling as a means of mining individuals' hidden emotions and psychological states from textual data to aid in theory development [4,5]. For instance, Jabr et al. [6] employed a thematic model to generate deeper insights into consumers' cognitive and emotional responses from product reviews. In extracting topics from textual data, researchers regard a topic model's interpretability, which captures the ease of comprehending and leveraging topics identified from texts, as a critical barometer of its quality because of its subsequent role in theory development [7,8]. Unlike other techniques of bolstering interpretability, such as interactive topic modeling [9] and topic labeling [10], embedding prior domain knowledge—which encapsulates the expertise and information already present in a select field into topic models—holds distinct advantages. These advantages stem from the targeted application of expert insights and structured knowledge to inform the topic model's learning process. Particularly, prior domain knowledge can yield topics intricately tied to domain-specific questions, allowing the customization of the topic model to fulfill specific domain needs and reducing the need for overly complex models or excessive parameter tuning, both of which are harder to achieve through other techniques [11]. Hence, scholars have increasingly called for embedding *a priori* domain knowledge within topic models to deal with the challenge of inadequate interpretability [11,12].

Yet, despite the growing prevalence of past studies that heed the call of scholars to integrate prior domain knowledge into text mining [11–13], their adaptability is limited to static domain knowledge that remains invariant over time. This is because previous attempts at integrating domain knowledge rely heavily on the availability of expert-curated vocabularies or pre-existing knowledge databases. This contrasts with reality because the rapid production of user-generated data often culminates in either (1) dynamic knowledge domains where expert-curated vocabularies or pre-existing knowledge databases rapidly become obsolete, or (2) new domains which are devoid of such resources entirely. In light of these constraints, this study explores how to automatically extract comprehensive domain knowledge from vast amounts of user-generated data, optimize the embedding of prior domain knowledge into topic models, and improve the interpretability of the resulting topic models. Several techniques for embedding domain knowledge into topic models, in the likes of anchored correlation

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explanations [13], knowledge graph-enhanced models [14], and supervised topic models [15], are well-documented within extant literature. However, these methods often overlook the importance of ensuring high-quality domain knowledge and are hindered by their complexity in implementation. To this end, we subscribe to the CorEx model as an appropriate technique for embedding domain knowledge into topic models that can circumvent the implementation complexity of traditional techniques by accommodating minimal domain knowledge and directly improving the probability distribution through anchor words [13,15]. Additionally, the quality and relevance of the topics generated by the CorEx model depend heavily on the selection of appropriate anchor words, which represent domain knowledge. Poorly chosen anchor words can lead to biased or less meaningful topics, ultimately reducing interpretability. Because the CorEx model relies on the modeler to provide anchor words based on domain knowledge or assumptions about the data, we further introduce a method for automatically generating anchor words. This enhancement optimizes the CorEx model's ability to embed domain knowledge and improves the interpretability of topic modeling.

Confronted with the necessity of enhancing model interpretability [8] and improving the technique of embedding domain knowledge, we advanced a novel LDA-GMM-CorEx topic model with prior domain knowledge to circumvent the limitations of contemporary topic modeling techniques. We first employed the traditional latent Dirichlet allocation (LDA) model based on a variational inference expectation-maximization (VI-EM) solution to conduct unsupervised topic modeling. Following this, the results of LDA were vectorized and input into a Gaussian mixture model (GMM) for clustering, and an LDA-GMM hybrid model automatically generated anchor words that integrate domain knowledge. On this basis, the CorEx topic model was deployed to generate topics. In other words, our model synergizes the complementary strengths of LDA, the GMM, and CorEx to address weaknesses arising from each method's standalone application. By combining these methods, our model bolsters the granularity and robustness of topic detection and interpretation in ways that are not possible when these methods are deployed independently.

More specifically, LDA offers a probabilistic framework for identifying thematic structures in textual data by capturing word co-occurrence patterns. However, LDA struggles with polysemy and synonymy, limiting its interpretability. The GMM enhances LDA by incorporating a continuous probability distribution to model word frequencies within topics, allowing finer distinctions between semantically similar but contextually different terms. CorEx adds an information-theoretic layer to the model, employing anchor words to steer the topic modeling process. This approach not only enhances the model's ability to discover more coherent and relevant topics but also increases interpretability by aligning topics more closely with domain-specific knowledge—a key aspect often lacking in purely statistical approaches like LDA and GMMs.

The combination of these three methods gives rise to a robust topic modeling framework that synergizes the probabilistic topic identification of LDA with the distributional flexibility of GMMs, and the information-theoretic benefits of CorEx. This combination addresses the semantic and statistical challenges in topic modeling, supplying a comprehensive solution that outperforms models relying on any of these methods in isolation.

We designed the LDA-GMM-CorEx topic model to effectively integrate prior domain knowledge and swiftly adapt to the rapid changes and unpredictability typical of digital data environments. We showcased the model's superior adaptability and enhanced interpretability to deliver crucial insights for decision-making. We applied this LDA-GMM-CorEx topic model to two online user-generated self-description corpora with differing characteristics to generate topics with prior domain knowledge. Finally, we assessed the semantic consistency and validity of the model.

The paper begins with an overview of contemporary topic modeling techniques and their application in IS research, followed by a discussion of their limitations. Section 3 describes the derivation of the LDA-GMM-CorEx topic modeling technique, elaborating on its structure, representation, and proposed extension. Next, we present two illustrative cases to demonstrate the application of the LDA-GMM-CorEx topic modeling approach to user-generated data elicited from Zhihu.com and Fiverr.com. By examining the semantic coherence and total correlation score of the topic models for these datasets, we summarize the results of our work and discuss their significance. Finally, we articulate the limitations of this study and suggest directions for future research.

2. Theoretical background

2.1. Topic models

Topic modeling is primarily deployed to uncover latent topic distributions in large text corpora. Originally introduced by Blei, Ng, and Jordan, LDA is a foundational technique that became pivotal in information retrieval, document classification, and content analysis [18]. As datasets become increasingly complex and massive in size, topic modeling methodologies have evolved to ensure scalability and applicability across various fields. Recent advances in disciplines such as computer science, information management, and marketing highlight the flexibility of these techniques, with each field customizing topic models to address its unique challenges and requirements.

In computer science, integrating deep learning with traditional topic models has led to significant improvements. For example, contextualized topic models (CTMs) utilize contextual embeddings from models like Bidirectional Encoder Representations from Transformers (BERT) to refine the accuracy and relevance of topics identified in large datasets. This approach is particularly beneficial for analyzing technical documents and online forums, where precise terminology is crucial [19]. In information management, researchers have adopted hierarchical Dirichlet processes to dynamically manage and categorize data. This model automatically adjusts the number of topics, making it ideal for large-scale database management and archival systems applications, where data volume and diversity require flexible and scalable solutions [20]. In marketing, aspect-based topic modeling (ABTM) plays a crucial role in extracting detailed insights from consumer feedback on social media and online review platforms. ABTM isolates specific product features or aspects mentioned by customers, giving marketers targeted insights into consumer preferences and behaviors [21].

In information systems (IS), researcher widely use traditional unsupervised topic models like Latent Semantic Analysis (LSA) and LDA because of their simplicity, low computational cost, and accessibility. These models operate under the text distribution assumption, suggesting that documents are generated by mixing multiple topics, with each topic characterized by a word distribution [22]. Many software libraries in Python and R [8] support these models, making them convenient for IS researchers to implement without requiring an additional Natural Language Processing (NLP) research background. LSA identifies the topic-based semantic relationship between text and words through matrix decomposition, but the factor load calculated by LSA has no clear interpretation, resulting in low interpretability [8]. Researchers developed LDA methods to improve this issue [18]. This method represents the topic of each document as a probability distribution.

Despite its advantages, LDA has significant limitations. One major drawback is topic overlap [7], where the same word appears n multiple topics. Therefore, the topics generated by LDA are not independent and orthogonal, diminishing their interpretability. Another limitation is LDA's inefficiency in handling complex text. This arises from its simplifying assumptions, such as treating documents as a mix of topics represented by word distributions. This approach can struggle with nuanced or specialized texts where context and the precise use of language are crucial, such as video bullet screen comments and personalized self-description texts. Additionally, LDA's "bag of words" approach

does not account for word order and context, making it difficult to capture the subtleties necessary for understanding complex texts. For example, in medical research papers, where terminology is dense and specific to certain subfields, LDA may fail to distinguish between topics related to different aspects of medicine [8]. Therefore, for complex topics, LDA often oversimplifies or misrepresents topic assignments, leading to inaccuracies when analyzing messy textual data [23].

2.2. Methodological research on topic modeling in information systems research

In this section, we summarize the literature on topic modeling in IS research [24] and list its research modes and specific topic modeling methods in Table 1. Although IS research frequently utilizes mainstream probabilistic models such as LSA and LDA for topic modeling, our literature analysis revealed a lack of diverse methodological research beyond these established models. Topic models like LDA and LSA have gained popularity as common text mining techniques because they efficiently provide an overview of topics within large text datasets [25]. Additionally, these models benefit from mature implementation tools and evaluation methods [8] that are easy for IS researchers to implement and that quickly identify potential semantic connections containing multiple topics [26]. However, traditional probability models struggle to meet the current demand for greater interpretability and adaptability, essential in fields requiring understanding and rapid responses to new data patterns.

To enhance the interpretability and adaptability of topic modeling, avoid topic overlap, and generate anchor-word-driven topics, IS scholars have tried various optimizations of traditional probabilistic models. For example, Nikolenko et al. [32] proposed an interval semi-supervised LDA (ISLDA) topic model to identify specific topics in qualitative research and assign specific words to those topics by fixing the Z-values of keywords associated with relevant topics. However, selecting anchor words is challenging because it requires significant effort to determine when words with multiple meanings appear. Andrzejewski et al. [11] proposed using logical constraints to enforce separability between topics, but this approach still does not solve the problem of how to anchor vocabulary. In addition, Jabr et al. [6] proposed improved LDA algorithms to identify various aspects of product reviews. Although this approach improves the interpretability of the topic model, it requires additional deletion and lexical extraction of the original corpus, potentially compromising the quality of the generated topics.

Table 1
Methodological research in information systems (IS) research.

Topics	Research Approach	Topic Modeling Method
Overview of LSI/LSA [27]	Model	LSA/LSA
Two hierarchical agglomerative clustering (HAC) techniques [28]	Model	HAC
Comparison of structured priors for LDA [29]	Comparative	LDA
Comparison of four topic modeling methods [4]	Comparative	LSA/PLSA/ LDA/CTM
Labeled LDA for tweet user characteristics [30]	Model	LDA
Methodological recommendations for LSA studies [31]	Model	LSA
Interval semi-supervised topic model (ISLDA) and coherence metric [32]	Model & Validation	ISLDA
Deconstructing reviews into aspects to evaluate the temporal evolution of user satisfaction with these aspects at a granular level [6]	Model	LDA
Using topic-labeled gold-standard sets to evaluate topic modeling interpretability [33]	Validation	LDA
Using a semi-automated approach to perform systematic literature reviews [34]	Model	LDA

2.3. Application research on topic modeling in is research

In this section we summarize the literature on applying topic modeling in IS research. We list its specific application scenarios and topic modeling methods in Table 2. According to our summary of the applied research literature, unsupervised probabilistic topic models such as LDA constitute the most commonly used method in IS. The probabilistic topic model's purpose is to extract the different aspects involved in the text. A common feature of topic models in this research is that they all need to extract topics from the corpora of specific events or domains. For example, we can learn about diabetes symptoms, diabetes diets, diabetes drugs, and clinical trials from threads in diabetes forums through topic modeling [35]. Moreover, it is necessary to extract the disaster-related aspects of tweets [36].

Based on the above literature, we discovered that the purpose of using topic modeling methods in most IS research is to explore the impact of digital content, and the role of topic modeling is largely descriptive in nature. Aspects or anchor words representing the underlying topic in the corpus of events or domains are called *prior domain knowledge*. Incorporation of domain knowledge into the basic topic model can be used to investigate embedded topics in textual data from different domains. This method directs the topic model, making it easier to explore how the text relates to a particular domain, such as disaster or service.

2.4. Topic-model embedded domain knowledge

Embedding domain knowledge into a topic model refers to integrating external expert knowledge or structured information about the domain into the model's learning process [11]. This incorporation can enhance the interpretability and applicability of topic modeling in specialized domains [11–13]. Researchers achieve this through various mechanisms, such as adjusting model priors, specifying relationships between words, or using annotated data to influence topic formation. Next, we explore various methodologies for embedding prior domain knowledge into topic models and discuss their respective advantages and disadvantages (Table 3).

Models such as Dirichlet forest priors [11] and anchored CorEx [13] provide mechanisms to guide topic discovery directly, based on word relationships or specific anchor words, enhancing the models' ability to

Table 2
Applied topic modeling in information systems (IS) research.

Focus	Purpose of Topic Modeling	Topic Modeling Method
Extract aspects from product reviews using multi-grain LDA [37]	Extract review aspects	MG-LDA
Review helpfulness and emotions shown in review [38]	Measure emotions	LSA
Call for use of LDA for theory generation [39]	Theory generation	LDA
Extract different aspects of satisfaction from reviews on shopping websites [6]	Extract review aspects	LDA
Learn whether Amazon reviews discuss different product aspects at different points in time [40]	Extract review aspects	LDA
Identify topics of natural disaster tweets on social media and manually summarize them [36]	Identify underlying topics	LDA
Analyze reactions in hotel reviews to the news that robots are replacing human services [41]	Extract perceptions of robots	LDA
Perform a systematic literature review using semi-automatic methods [34]	Identify underlying topics	LDA
Examine the determinants of the Top 500 companies [42]	Interpret factors	-
Identify topics related to diabetes from diabetes forums [35]	Identify underlying topics	LDA

Table 3
Methods Embedding Prior Domain Knowledge.

Method	Focus	Advantages	Disadvantages
Dirichlet Forest Priors [11]	Andrzejewski et al. introduced Dirichlet forest priors to encode domain knowledge in LDA models, allowing the specification of word relationships.	Enables the model to incorporate complex word relationships, improving topic coherence	Specification of Dirichlet Forests can be complex and requires significant domain expertise.
Anchored CorEx [13]	Allows users to specify anchor words to guide topic formation	Provides a straightforward way to incorporate domain knowledge through anchor words, enhancing interpretability	Selection of appropriate anchor words is crucial and can be subjective.
Supervised Topic Models (LDA) [15]	Extended LDA to incorporate supervision in the form of document-level labels or responses	Enhances predictive power of the model for supervised tasks, making the topics more task-oriented	Relies on the availability of labeled data, which may not be available for all datasets
Knowledge Graph-Enhanced Models [14]	Recent works have explored the integration of knowledge graphs with topic models to enrich the topics with structured semantic information.	Leverages rich, structured domain knowledge, potentially improving topic relevance and interpretability across different domains	Construction and integration of knowledge graphs add complexity to the modeling process.

generate coherent and interpretable topics. Supervised topic models and knowledge graph-enhanced models leverage external labels or structured semantic information to align the discovered topics more closely with specific tasks or broader domain contexts. Although these methods offer significant advantages, such as improved topic coherence and relevance, they also introduce challenges related to the availability and quality of domain knowledge, as well as increased model complexity.

The CorEx model [13] excels in incorporating prior domain knowledge through anchor words, enhancing the interpretability of the topics it generates. The CorEx model does not make any structural a priori assumptions about data generation. Instead, it uses an approach similar to information gain [43], relying on differences in total correlations to find the topic that best explains the correlation of the data. The introduction of anchor words allows minimal yet effective incorporation of domain knowledge, enhancing the interpretability and relevance of the resulting topics. Table 4 summarizes the merits of CorEx in embedding domain knowledge compared to other methods:

Our review of the literature reveals that past studies have made various attempts to incorporate expert insights and structured domain information into topic models to enhance their interpretability. However, adapting these topic models to interdisciplinary, novel domains, or domains where domain knowledge changes dynamically, remains a challenge. This is especially true in fields such as emerging technologies. Consequently, effectively generating high-quality domain knowledge is the key to optimizing topic models.

3. Development of the LDA-GMM-CorEx topic model

In this section, we advance the idea of a novel topic-modeling approach, labeled as the LDA-GMM-CorEx topic model because of its three components: the LDA topic model, the GMM, and CorEx. CorEx identifies topics that maximize total correlation with the data, capturing more complex structures than LDA. CorEx can generate a highly interpretive distribution of topics by anchoring words and information

Table 4
Comparison of CorEx with Other Methods of Embedding Domain Knowledge.

Advantages	Description
Ease of Incorporating Domain Knowledge [13]	CorEx allows the direct and intuitive incorporation of domain knowledge through anchor words. Users can specify words that they believe are central to a topic, and the model uses this information to guide the discovery of related topics. This process is more transparent and easier to understand and implement than methods requiring complex mathematical formulations or extensive preprocessing.
Discovery of Latent Structure beyond Anchors [16]	Although CorEx leverages anchor words to guide topic discovery, it also excels at uncovering latent structures and topics that may not be directly tied to the provided anchors. This capacity to reveal hidden structures enriches the analysis, offering insights beyond the initial domain knowledge.
Hierarchical Topic Modeling [17]	CorEx supports hierarchical topic modeling, enabling the discovery of topics at multiple levels of granularity. This hierarchical structure can provide deeper insights into the relationships between topics, offering a more nuanced understanding of the data. This aspect is particularly useful in complex domains where topics can be nested or interrelated in intricate ways.
Flexibility and Adaptability [13]	The model's flexibility allows users to experiment with different sets of anchor words with relative ease, adapting the model to various domains and types of textual data. This adaptability contrasts with models that might require extensive reconfiguration or fine-tuning when applied to new domains or datasets.
Minimal Required Knowledge [13,16,17]	Unlike methods that require extensive labeled data or complex structured knowledge (e.g., knowledge graphs), CorEx can benefit from minimal domain knowledge. A small set of anchor words is sufficient to significantly influence the model's outcomes, making it particularly appealing in scenarios in which detailed domain knowledge is sparse or hard to encode.

bottlenecks [13]. An important feature of the CorEx model for incorporating domain knowledge is its support for anchoring, which allows users to guide the topic discovery process by specifying certain variables or topic words that should be associated with specific underlying factors.

However, a key challenge with CorEx is generating high-quality anchor words automatically instead of relying on manual selection. The process of automatically generating anchor words that represent the domain knowledge has yet to be explored. The combination of LDA and the GMM can solve this issue effectively. Specifically, LDA is well suited to large datasets and offers interpretable results, which are crucial for the initial exploration of the text data. It helps identify a broad set of potentially relevant words for topics by capturing the most prominent topic based on word co-occurrence. These words can serve as initial candidates for anchor words in CorEx. The GMM can refine this initial selection of anchor words by analyzing the subtleties in word usage and distribution in feature space, often revealing finer distinctions or context-specific usages. Specifically, it can discern subtle similarities and differences in the usage of words based on their distribution in a transformed feature space (e.g., TF-IDF). This capability allows finer distinctions among words that might appear similar through LDA but which actually belong to different contexts or usages, thus refining the selection of anchor words. The GMM's probabilistic nature manages topic overlap better, providing a more flexible model of word associations.

Using anchor words selected and refined by LDA and the GMM, CorEx constructs topics that are statistically significant and highly relevant to specific inquiries or known structures within the dataset, enhancing topic specificity. This method allows controlled data

exploration, focusing on particular areas of interest or importance in the dataset. This combination leverages both the probabilistic strengths of LDA and the GMM and the information-theoretic capabilities of CorEx, creating a robust model that adapts to broad patterns and intricate dataset details.

First, we used an integrated algorithm of LDA and the GMM to automatically generate anchor words representing domain knowledge. LDA is a topic modeling technique that discovers the underlying topic structures in a collection of documents, whereas the GMM is a clustering technique that models data as a mixture of Gaussian distributions. The integration focuses on converting and combining the parameters and outputs of LDA into inputs suitable for GMMs. Using VI-EM LDA topic modeling, we efficiently identified multiple topic words and their distributions across numerous documents [44].

3.1. Combining LDA and GMM for anchor word selection

LDA applies probabilistic cryptic meaning analysis based on the Bayesian inference mechanism [45], and Dirichlet distribution is a probability distribution of multiple continuous random variables. Using LDA for topic analysis requires inferring all model parameters by estimating a posterior probability distribution for a given text set to learn the topic distribution of each text and the word distribution of each topic. LDA assumes each topic will correspond to a lexical distribution and each document will correspond to a topic distribution. By analyzing the distribution of topics in the document, LDA can give each topic a probability distribution to conduct topic clustering or text classification.

LDA assumes that the prior distribution of the topic of a document is a Dirichlet distribution; that is, for any document $m \in \{1, ..., M\}$, the topic distribution vector ϑm is $\vartheta m \sim$ Dirichlet (α), where α is both a hyperparameter of the distribution and a K-dimensional vector. The Dirichlet prior distribution of the hyperparameters α is as follows [46]:

$$P(\vartheta|\alpha) = \frac{\Gamma\left(\sum_m \alpha_m\right)}{\prod_m \Gamma(\alpha_m)} \prod_m \vartheta^{\alpha_m - 1}. \tag{1}$$

LDA assumes that the prior distribution of words in a topic is also a Dirichlet distribution; that is, for any topic $k \in \{1, ..., K\}$, its word distribution φ_k is $\varphi_k \sim$ Dirichlet (β), where β is the hyperparameter of the distribution and also a V-dimensional vector. V represents the number of words in the vocabulary. The Dirichlet prior distribution of the hyperparameters β is as follows [46]:

$$P(\varphi|\beta) = \frac{\Gamma\left(\sum_k \beta_k\right)}{\prod_k \Gamma(\beta_k)} \prod_k \varphi^{\beta_k - 1}. \tag{2}$$

For the Nth word in any document d in the data, we can determine the distribution of its topic number z_{mn} from the topic distribution ϑm: $z_{mn} = \text{multi}(\vartheta m)$. For this topic number, the probability distribution of observable words ω_{mn} is $\omega_{mn} = \text{multi}(\beta z_{mn})$.

LDA defines a joint probability distribution of observed and hidden variables as follows:

$$P(\omega, z, \vartheta, \varphi|\alpha, \beta) = P(\vartheta|\alpha)P(\varphi|\beta)P(z|\vartheta)P(\omega|z, \varphi). \tag{3}$$

However, this does not explain how the inference is performed. In this research, we use the VI-EM method to perform this inference, estimating the distribution of topics in a document and the distribution of words in topics.

The EM algorithm is a general technique for finding maximum likelihood estimates of parameters in probabilistic models, particularly when the model includes latent variables. EM alternates between two steps:

- The expectation (E) step calculates the expected value of the log-likelihood function with respect to the conditional distribution of the latent variables given the observed data and current parameter estimates.
- The maximization (M) step finds the parameters that maximize the expected log-likelihood calculated in the E step.

EM is iterative and continues until convergence. However, directly applying the standard EM algorithm to LDA is impractical. The LDA model has hidden variables z, ϑ, φ, and the parameters of the model are α, β. To find the model parameters and the corresponding hidden variable distributions, the EM algorithm needs to find the conditional probability distribution expectation of the hidden variables z, ϑ, φ in step E and maximize this expectation in step M to get the updated posterior model parameters α, β. However, in the E step, the interdependence between the θ, β, z components—which is called coupling—makes it difficult to find the conditional probability distribution of the hidden variables θ, β, z, and the corresponding expectation. To address this, we introduced the VI method.

VI approximates the true posterior distribution in Bayesian models when it is computationally difficult to calculate accurately. VI turns the inference problem into an optimization problem, in which the goal is to find a simpler (variational) distribution that is close to the true posterior distribution. The method assumes that the hidden variable θ is formed by the independent distribution γ, z by η, and φ by ρ. By using the variational distribution, VI stimulates the posterior probability distribution of the approximate hidden variable, removing the coupling relationship and enabling the EM algorithm to be performed. The variational distribution q is expressed as follows:

$$q(\vartheta, z, \varphi|\gamma, \rho, \eta) = \prod_{k=1}^{K} q(\varphi_k|\rho_k) \prod_{d=1}^{M} q(\vartheta_d|\gamma_d) \prod_{n=1}^{N} q(z_n|\eta_n). \tag{4}$$

Then it uses Kullback–Leibler divergence (KL) to calculate the similarity between the two. Our goal is to find the parameters γ and η that minimize KL divergence. The KL divergence formula is as follows:

$$\text{KL}(q(\vartheta, z, \varphi|\gamma, \eta, \rho)\| \, p(\vartheta, z, \varphi|w, \alpha, \beta)) = \text{Eq}\left(\log \frac{q(\vartheta, z, \varphi|\gamma, \eta, \rho)}{p(\vartheta, z, \varphi|w, \alpha, \beta)}\right) = -(\text{Eq}[\log(p(\varphi, \theta, z, \text{w}|\alpha, \beta))] - \text{Eq}[\log(q(\varphi, \ \theta, z|\rho, \gamma, \eta))]) + \log(p(w)). \tag{5}$$

Let $\mathscr{L}(\rho, \gamma, \eta, \alpha, \beta)$ as $\text{Eq}[\log(p(\varphi, \theta, z, w|\alpha, \beta))] - \text{Eq}[\log(q(\varphi, \ \theta, z|\rho, \gamma, \eta))]$, which is named ELBO (evidence lower bound). Because log (p (w)) is a constant value, minimizing $\text{KL}(q(\vartheta, z|\gamma, \eta) \parallel p(\vartheta, z|w, \alpha, \varphi))$ requires maximizing $\mathscr{L}(\gamma, \eta, \alpha, \varphi)$. Expanding L according to Bayes yields the following formula:

$$\begin{aligned}\mathscr{L}(\rho, \gamma, \eta, \alpha, \beta) = {} & \text{Eq}[\log(p(\varphi|\beta))] + \text{Eq}[\log(p(\text{z}|\vartheta))] + \text{Eq}[\log(p(\vartheta|\alpha))] \\ & + \text{Eq}[\log(p(w|\varphi, z))] - \text{Eq}[\log(q(\varphi|\rho))] - \text{Eq}[\log(q(z|\eta))] \\ & - \text{Eq}[\log(\text{q}(\vartheta|\gamma))].\end{aligned} \tag{6}$$

By solving for these seven terms of ELBO, we can finally obtain ELBO's expression for the variational parameters ρ, γ, η. Take the first expression Eq[log(p(φ|β))], for example:

$$\mathrm{Eq}[\log(p(\varphi|\beta))] = K\log\Gamma\left(\sum_{i=1}^{V}\beta_i\right) - K\sum_{i=1}^{V}\log\Gamma(\beta_i) + \sum_{k=1}^{K} \times \sum_{i=1}^{V}(\beta_i - 1)\left[\Psi(p_{ki}) - \Psi\left(\sum_{i'=1}^{V}p_{ki'}\right)\right] \quad (7)$$

where $\psi(x) = \frac{d}{dx}\log\Gamma(x) = \frac{\Gamma'(x)}{\Gamma(x)}$.

Then we can execute the EM algorithm to obtain these expressions. The goal of step E is to obtain the optimal variational parameters. We obtain an iterative expression by differentiating each variational parameter λ, ϕ, γ of the ELBO function and making the partial derivative 0. We can obtain the best variational parameter after multiple iterations of convergence:

$$\gamma_k = \alpha_k + \sum_{n=1}^{N}\eta_{nk} \quad (8)$$

$$\rho_{ki} = \beta_i + \sum_{m=1}^{M}\sum_{n=1}^{N_m}\eta_{mnk}w_{mn}^{v} \quad (9)$$

$$\eta_{nk} \propto \exp\left(\sum_{i=1}^{V}w_n^i\left[\Psi(p_{ki}) - \Psi\left(\sum_{i'=1}^{V}p_{ki'}\right)\right] + \Psi(\gamma_k) - \Psi\left(\sum_{k'=1}^{k}\gamma_{k'}\right)\right). \quad (10)$$

Then, we iterate repeatedly until the three variational parameters converge.

The goal of the M step is to obtain the optimal model parameters by fixing the variational parameters and maximizing ELBO. The iterative formula of the model parameters is as follows:

$$\alpha_k = \alpha_k + \frac{M\left[\Psi\left(\sum_{k'=1}^{k}\alpha_{k'}\right) - \Psi(\alpha_k)\right] + \sum_{d=1}^{M}\left[\Psi(\gamma_{dk}) - \Psi\left(\sum_{k'=1}^{k}\gamma_{dk'}\right)\right]}{M\left[\Psi'\left(\sum_{k'=1}^{k}\alpha_{k'}\right) - \delta(k,j)\Psi'(\alpha_k)\right]}, \quad (11)$$

where δ (*k*, *j*) = 1 only if $k = j$; otherwise, δ (*k*, *j*) = 0.

$$\beta_i = \beta_i + \frac{K\left[\Psi\left(\sum_{i'=1}^{V}\beta_{i'}\right) - \Psi(\beta_i)\right] + \sum_{k=1}^{K}\left[\Psi(\rho_{ki}) - \Psi\left(\sum_{i'=1}^{V}\rho_{ki'}\right)\right]}{K\left[\Psi'\left(\sum_{i'=1}^{V}\beta_{i'}\right) - \delta(i,j)\Psi'(\beta_i)\right]}, \quad (12)$$

where δ (*i*, *j*) = 1 only if $i = j$; otherwise, δ (*i*, *j*) = 0.

After completing the algorithm, we can get the posterior parameters of the model α, β; the approximate model topic word distribution ρ; and the approximate training document topic distribution γ.

In our approach, VI provides a tractable approximation of this posterior, making the E step computable. We combined VI and EM in a VI-EM framework to estimate the LDA model parameters. This approach uses VI to perform the E step, approximating the posterior over the latent variables (topic assignments for each word and topic distributions for each document) with a simpler distribution. The M step maximizes the expected log-likelihood with respect to the model parameters (topic-word distributions), as approximated by VI. The combination of VI and EM leverages the strengths of both approaches to efficiently estimate the parameters of complex probabilistic models.

Next, we applied the GMM to the document's topic distribution vectors to cluster them based on topic composition to generate anchor words automatically for each topic. The GMM is a probabilistic model that assumes data points are generated from a mixture of a finite number of Gaussian distributions with unknown parameters. It clusters data points to find which Gaussian distribution (or "cluster") a point most likely belongs to. The posterior probability of each data point indicates that it belongs to various types of possibilities, rather than only determining that it belongs to a certain class, which is why it is called "soft clustering." Specifically, the model uses multiple Gaussian distribution functions to approximate the probability distribution of any shape. Hence, The GMM is composed of multiple single Gaussian density distributions, each of which is called a "component," and their linear sum forms the probability density function of GMMs. This approach regards data points to be clustered as distributed sampling points. and estimates the parameters of a Gaussian distribution using a method similar to maximum likelihood estimation. It applies the EM algorithm to solve for these parameters, obtaining the membership function for classifying data point pairs.

We define a GMM as a weighted sum of *M* component Gaussian densities:

$$p(x|\lambda) = \sum_{i=1}^{M}\omega_i p(x|\lambda)\mathrm{N}\left(x\middle|\mu_i, \sum i\right), \quad (13)$$

where x represents a data point (in our case, x is the keyword represented by its features, such as LDA topic distributions), λ represents the GMM parameters, ω_i are the mixture weights, $\mathrm{N}(x|\mu_i, \sum i)$ is the component Gaussian density, $\sum i$ are the mean and covariance, and M is the number of components.

We use the EM Algorithm to find the maximum likelihood estimates of the parameters. The algorithm proceeds iteratively in two steps: the expectation (E) step and the maximization (M) step. In the E step, we calculate the responsibility that component k has for data point x:

$$\gamma(Z_{nk}) = \frac{\pi_k \mathrm{N}(x_n|\mu_k, \sum k)}{\sum_{j=1}^{M}\pi_j \mathrm{N}\left(x_n\middle|\mu_j, \sum j\right)}. \quad (14)$$

The goal of the M step is to update the parameters using the responsibility calculated in the E step. Specifically, we obtain the formulas for μ and $\sum$ by differentiating the expected data log-likelihood with respect to the particular parameters and setting the derivatives to zero:

$$\mu_k = \frac{\sum_{n=1}^{N}\gamma(Z_{nk})x_n}{\sum_{n=1}^{N}\gamma(Z_{nk})} \quad (15)$$

$$\sum k = \frac{\sum_{n=1}^{N}\gamma(Z_{nk})(x_n - \mu_k)(x_n - \mu_k)^T}{\sum_{n=1}^{N}\gamma(Z_{nk})}, \quad (16)$$

and then derive the formula for π by maximizing the log-likelihood:

$$\pi_k = \frac{1}{N}\sum_{n=1}^{N}\gamma(Z_{nk}). \quad (17)$$

As a consequence, the LDA-GMM method can maximize and quickly acquire merged anchor words, such as "hobby," "game," and "health," which represent underlying topics in the corpus and are therefore also referred to as domain knowledge.

3.2. Integration with the CorEx model

After preparing the anchor words, we input them into the domain knowledge-driven CorEx topic model. The core principle of this step is to improve the topic word probability distribution by integrating anchor words with domain knowledge before the CorEx model is implemented.

The CorEx model is an information-theoretic approach designed to uncover meaningful data structures, especially when latent factors are discovered that explain the relationships and dependencies between variables. It is particularly adept at handling large, complex datasets with which traditional methods might struggle. The core idea behind CorEx is to maximize the total correlation (also called "multivariate mutual information") between the observed variables and the latent factors. Total correlation measures how much information is shared among variables; it goes beyond pairwise correlations to capture more complex relationships.

First, we used the form of KL divergence to define total correlation (TC) or multivariate mutual information [47] as follows:

$$\mathrm{TC}(X_G) = \sum_{i\in G}H(X_i) - H(X_G) = D_{\mathrm{KL}}\left(P(X_G)\|\prod_{i\in G}P(X_i)\right), \quad (18)$$

where $H(X)$ is the entropy of the discrete random variable X, and G is a subset of X. In simple terms, it is the sum of the entropies of each random variable in the subset minus their joint entropies. TC can be regarded as the KL divergence between the joint and edge distribution multiplicative [43]. Thus, when TC is zero, KL divergence is also zero, and the joint distribution is equal to the edge distribution multiplicative, meaning there is no correlation, and the variables are independent.

Conditional TC is defined as follows:

$$\mathrm{TC}(X|Y) = \sum_{i \in G} H(X_i|Y) - H(X|Y). \tag{19}$$

Using this, we can measure how much Y explains the correlation of X:

$$\mathrm{TC}(X; Y) = \mathrm{TC}(X_G) - \mathrm{TC}(X|Y). \tag{20}$$

If $\mathrm{TC}(X|Y)$ is equal to zero, $\mathrm{TC}(X; Y)$ will take the maximum value, indicating that Y explains all the correlations of X. To maximize the document's TC value and improve topic representation, the CorEx algorithm restarts multiple times and performs the run with the highest TC value.

In contrast to previous practice, this paper uses anchor words that are generated by the LDA-GMM model rather than by the researcher. Merged anchor words can enhance the document topic distribution by increasing the prominence of topics related to the anchor words it contains [12]. Furthermore, by setting the anchoring intensity, this method changes how the anchoring words affect the topic model. Anchor strength controls how much weight the CorEx topic model maximizes on the multivariate mutual information between the anchoring words and their respective topics [13]. The higher the intensity, the greater the influence of anchor words on the topic model. Generally, when anchor strength is set to a value less than five, the effect is not mandatory but encourages topic word distribution to be generated according to the topic related to the anchor word. For example, we can anchor "doctor," "teacher," and "programmer" to one topic, and anchor "medicine," "computer science," and "economics" to another, helping to separate and merge discussions of "career" and "major" topics. We describe this whole process by illustrating several key steps.

3.3. Implementation of LDA-GMM-CorEx model

Fig. 1 shows the graphical structure of the LDA-GMM-CorEx topic model. We then show how we connected the three components to develop the LDA-GMM-CorEx model. The first step involves conducting topic modeling with VI-EM LDA. We used VI-EM LDA to discover potential topics in the document collection. For each document, VI-EM LDA generates a topic distribution vector θ, representing the proportion of each topic in the document. This vector comes from a Dirichlet distribution parameterized by α, and each element θ indicates the presence or proportion of a topic in the document. For document d, this vector is represented as θ_d, where each element corresponds to a topic.

Then, we needed to transform the outcome of VI-EM LDA, θ, into a form suitable for GMM input. Specifically, we used the pretrained Word2vec[1] model provided by HanLP to obtain word embedding. This model utilizes combined skip-grams and negative sampling techniques and combines binary and character-level information to generate a 300-dimensional embedding of words, which can capture the semantic and syntactic nuances of the language. Each word in the output of LDA is represented as a dense vector in a continuous vector space. The generated vectors encapsulate the semantic features of the outcome words of the VI-EM LDA and thus serve as inputs to the GMM. This approach combines the semantic richness of word embedding with the flexibility of GMMs in modeling data distributions.

[1] The link for word2vec is https://hanlp.hankcs.com/docs/api/hanlp/pretrained/word2vec.html.

Transitioning from LDA's topic distributions to Word2Vec's word embeddings can lead to a loss of thematic information. Although Word2Vec captures detailed semantic relationships, it may not align these relationships inherently with the broader topics identified by LDA. This misalignment can lead to semantic inconsistency, in which the rich thematic insights provided by LDA are not effectively reflected in the dense semantic vectors produced by Word2Vec. To ensure that the topic distributions from LDA are effectively aligned with the semantic vectors generated by Word2Vec, we used the topic-weighted Word2Vec model, in which each word vector is weighted according to its probability distribution under topics derived from LDA, thereby preserving thematic relevance.

We first computed embedding for each topic by identifying the top N words based on their probability distribution within the topic. This step focuses on the words most representative of each topic's semantic field. Then, we retrieved word embeddings by using a well-validated, pretrained Word2Vec model trained on a large and domain-relevant corpus to retrieve embeddings for these top words. The choice of a pretrained model allowed leveraging a rich semantic space developed on large, diverse datasets and captured a wide array of syntactic and semantic word relationships. After this step, we created an embedding for each topic by averaging the embeddings of its top N words, weighted by their probability within the topic. This process resulted in a dense vector that captured the semantic essence of each topic. This approach aligns with established practices in the field. Studies by Mikolov [48] provided a foundation for using word embeddings to enhance topic model outputs. Additionally, research on integrating mixed membership models like LDA with embeddings [49] supports the validity of this methodology.

To address the potential alignment issues between LDA-derived topics and GMM-based semantic clustering, we conducted additional experiments to assess the stability and consistency of these alignments using silhouette scores. They measure how similar an object is to its own cluster compared to other clusters, with higher scores indicating better-defined clusters. The evaluation results are shown in Table 5:

Results demonstrate that the topic-weighted Word2Vec approach significantly improved the cohesion and separation of semantic clusters in the GMM compared to standard Word2Vec embeddings. This methodology ensures that the semantic integrity of the topics is preserved in the document embeddings. By using weighted averages of word embeddings, it maintains a close connection between the original textual data and the topics identified by LDA.

Next, we used the converted word vectors as the GMM's observable data, X_i, and conducted clustering with GMM. Fitting the GMM to word vectors models the data distribution for cluster analysis of similar words. Z_n represents the result of clustering the keywords, indicating the cluster from the mixture to which each X_i most likely belongs to. In the clustering context, the model infers the values of Z_n based on the observed X_n, as well as the parameters of each Gaussian component in the mixture (represented by μ and Σ) and the mixing coefficients (π). We obtained the parameters of each Gaussian distribution after the EM algorithm converged. We then assigned each keyword to the cluster with the highest probability represented by a Gaussian distribution, producing keyword clusters.

After this, we selected anchor words. From each Gaussian cluster, we chose representative words based on proximity to the cluster center. During this selection process, we ensured that the selected anchor words were relevant and had a clear semantic meaning to improve the interpretability of the CorEx topics.

Finally, we incorporated anchor words into the CorEx model. In CorEx, anchor words can be used to steer the model toward discovering topics aligned with their semantic significance. The CorEx topic model defines a topic by $P(Y|X)$, which represents a discrete random variable that can affect X, with a value range of k possibilities. During initialization, we randomly set α and the topic distribution $P(Y|X)$ of the documents. Semi-supervised learning is achieved by fixing some values of the connectivity matrix. Normally, α is in the (0, 1) interval, and

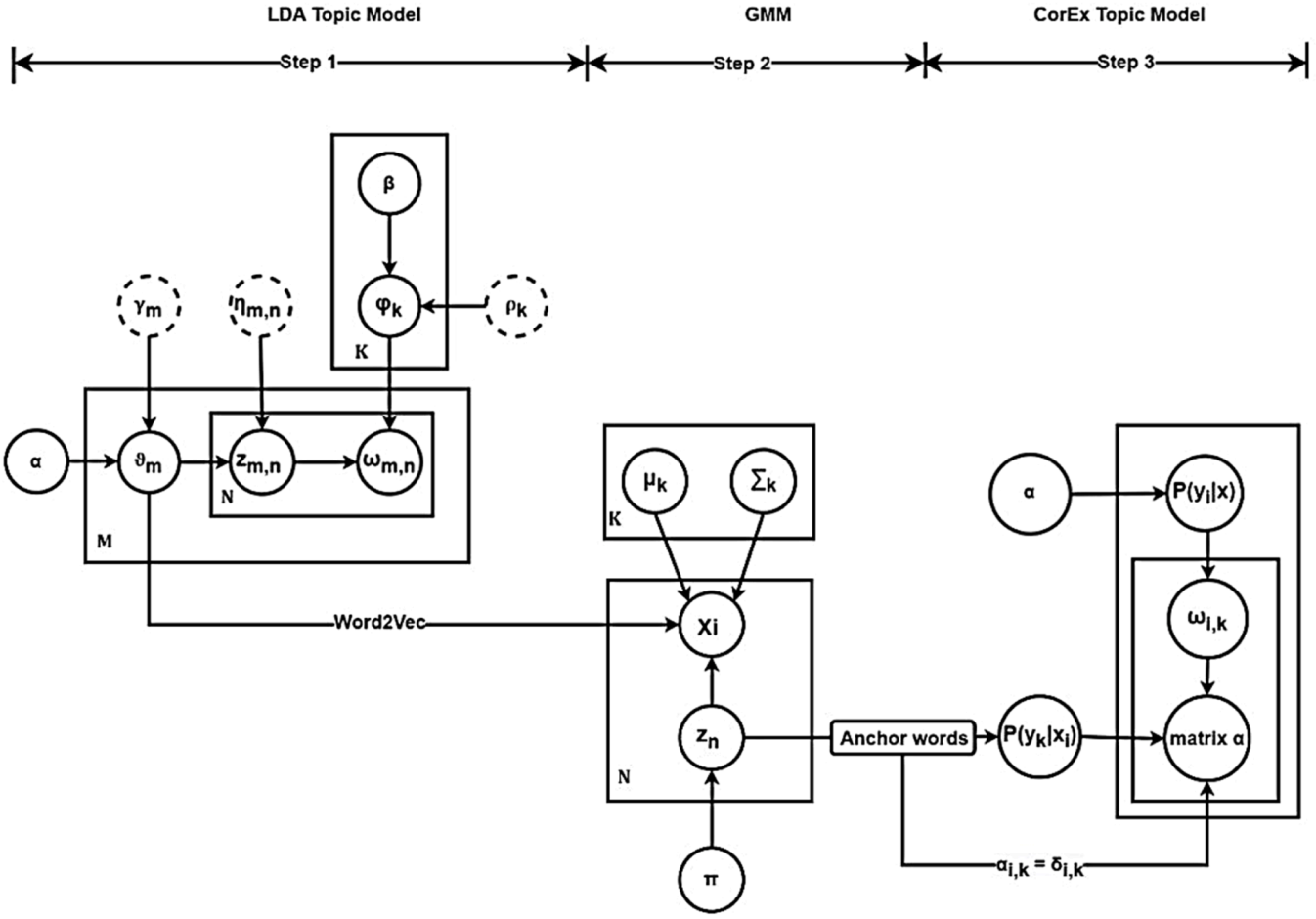


Fig. 1. Graphical representation of the LDA-GMM-CorEx model.

Table 5
Silhouette Score for Two Corpus.

Silhouette Score	Zhihu.com	Fiverr.com
Standard Word2Vec	0.0343	0.1144
Topic-Weighted Word2Vec	0.7344	0.1387

Table 6
Methodological research in information systems (IS).

Parameters	Definition
α	Parameter for the Dirichlet prior on the per-document topic distributions
θ_m	Topic distribution for document m
ϕ_k	Word distribution for topic k
β	Parameter for the Dirichlet prior on the per-topic word distribution
$z_{m,n}$	Topic for the nth word in document m
$\omega_{m,n}$	Specific word
$\eta_{m,n}$	Number of times the nth term in document m is assigned to topic k
Σk	Covariance matrix for cluster k in the mixture model
μ_k	Mean of cluster k in the mixture model
π	Mixing coefficient for the clusters
X_i	Data point that is being clustered
Z_n	Cluster assignment for x_i
α	Matrix assigned to subsets of word topics
$P(y_i\|x)$	Topic distribution of the document
$P(y_k)$	Probability of each topic
Matrix α	Matrix of the regularization parameters
$P(y_k\|x_i)$	Topic distribution of each word
$w_{i,k}$	Strength of association between feature x_i and latent factor y_k
δ_i	Strength of the anchor

anchoring the word i to topic k makes $\alpha_{i,k} = \delta_{i,k}$, where δ is the anchoring strength [13]. After we input the anchor words into the CorEx model, anchoring word i to topic k can make $\alpha_{i,k} = \delta_{i,k}$. Then the CorEx topic model continuously updated the probability of each topic $P(y_k)$, the topic distribution $P(y_k|x_i)$ of each word, the distribution matrix α of words to topic subsets, and the topic distribution $P(y_k|x)$ of each document according to the formula. Using LDA-GMM-derived anchor words should lead to topics that are more interpretable and relevant to the underlying clusters in the data. We list all the parameters in the LDA-GMM-CorEx model and their corresponding meanings in Table 6:

Each of these steps describes a progression from simple topic modeling with LDA, which assumes that topics are mixed in a simple way within documents, to the GMM, which models data points as coming from multiple Gaussian distributions, to CorEx, which seeks to model not just data point distributions but also their correlations.

3.4. Advantages of the LDA-GMM-CorEx model

Our hybrid model—which integrates LDA, GMM, and CorEx—offers several distinct advantages. First, the hybrid model enhances adaptability by adjusting to changes in data characteristics, such as shifts in vocabulary or emerging topics. This adaptability is facilitated by the flexibility of GMMs in handling diverse data distributions and CorEx's ability to update anchor words as new knowledge becomes available.

Second, our model significantly improves interpretability. By integrating CorEx, we directly incorporate domain expertise into the modeling process. This integration results in topics that are not only statistically significant but also meaningful and actionable from a domain perspective. The anchor words serve as semantic signposts,

enhancing the coherence and relevance of the topics generated.

Third, the introduction of domain-driven anchor words via CorEx, combined with the statistical power of LDA and the flexibility of GMMs, offers a novel framework. This framework is particularly beneficial in complex domains that require both broad and deep text analysis, optimizing the balance between mathematical robustness and practical usability.

Each component of the model contributes uniquely to the handling of large datasets. LDA is efficient at identifying overarching themes by analyzing word co-occurrences and distributions across documents. This is particularly valuable in large datasets in which main themes may not be immediately apparent. The GMM complements LDA by modeling word distributions as mixtures of Gaussian distributions rather than discrete distributions. This allows a more nuanced understanding of word usage, capturing the frequency and variation of words within topics and providing a more refined topic structure. CorEx enhances the model by introducing an information-theoretic approach to topic modeling. By using specific anchor words based on domain knowledge, CorEx guides the topic discovery process toward more relevant and coherent themes, essential in large datasets in which data volume can overwhelm traditional models.

Finally, in terms of computational efficiency, the synergy between these methods allows for scalable yet detailed analysis. LDA provides baseline efficiency, the GMM adds flexibility for handling continuous data characteristics, and CorEx improves topic coherence and ensures alignment with domain-specific knowledge without significantly increasing the computational burden.

As a consequence, our proposed topic model enables a deeper and more targeted exploration of text data while improving the applicability and interpretability of the results, making it a powerful strategy for advanced topic modeling tasks in large datasets.

4. Illustrative cases

4.1. The choice of two illustrative cases

The performance and effectiveness of topic models, such as LDA, are influenced by factors like the language of the text and document length. These aspects significantly affect the quality of topics extracted from a corpus due to differences in language morphology, grammatical structures, and the availability of language processing tools. Additionally, short texts may not provide enough contextual information for effective topic modeling because of sparse word co-occurrences. To demonstrate the versatility and robustness of our proposed LDA-GMM-CorEx topic modeling method across various text types, we conducted illustrative case studies using datasets from Zhihu.com and Fiverr.com. The Zhihu.com dataset primarily consists of short-text user profiles in Chinese, whereas the Fiverr.com dataset includes longer service-related text descriptions in English. These choices allow a comparative analysis of how our model performs with different text lengths and complexities and provide insights into its cross-linguistic capabilities, ensuring global applicability. Additionally, we present detailed statistical data on text length (number of words) for each platform to address variability within the datasets. The descriptive statistics for the Zhihu.com and Fiverr.com datasets, highlighting notable differences in sample size, text length, language, and domain, are shown in Table 7:

Table 7
Descriptive statistics.

Characteristics	Zhihu.com	fiverr.com
Sample size	4,290,484	7,491
Language	Chinese	English
Mean Text Length	8.364	435.997
Median Text Length	6.000	365.000
S.D. of Text Length	7.371	304.390
Domain	Personal information	Service-related information

The Zhihu.com dataset is significantly larger, containing over 4.29 million entries written predominantly in Chinese and primarily related to personal information. These entries are notably brief, with a mean text length of 8.364 words, a median of 6 words, and a low standard deviation of 7.371, suggesting little variation in text length. In contrast, the Fiverr.com dataset is much smaller, with 7,491 entries in English, consisting mainly of service-related descriptions. The texts in this dataset are considerably longer, averaging 435.997 words per entry, with a median of 365 words and a high standard deviation of 304.390, indicating wider variation in text length.

In our first case study, we analyzed over 4.29 million user-generated personal descriptions from Zhihu.com, a major Chinese online Q&A platform. We selected this dataset not only for its size but also because it represents a less-studied domain in topic modeling, often associated with challenges such as low interpretability and topic overlap. We demonstrated the model's ability to improve topic separation and interpretability by using LDA for initial topic word extraction, constructing anchor words with GMMs, and embedding these into the CorEx model.

Our second case study involved analyzing 7,491 service-related descriptions from Fiverr.com, a prominent online marketplace for freelance services. This English dataset provided a contrast to the Zhihu.com corpus, showcasing the model's efficacy in handling longer texts and its adaptability to different linguistic and contextual demands. We employed VI-EM LDA to extract topics and used GMM-derived anchor words to enhance the CorEx model, further illustrating our method's strengths in a diverse textual environment.

By comparing the results from these distinct datasets, we validated our model's effectiveness across different languages and document lengths, highlighting its potential in diverse IS applications.

4.2. Data collection and pre-processing

To ensure comparability across models, all textual data underwent the same pre-processing pipeline, including tokenization, lemmatization, stopword removal, and constructing a term-document matrix. This standardized preparation ensures that the input data for each model is comparable.

4.2.1. Chinese short text corpus from Zhihu.com

We collected personal disclosure text information (Fig. 2) from Zhihu.com, the largest Chinese online question-and-answer community. To obtain as much user data as possible, we used a recursive crawl method. Specifically, we started by finding a user with many followers. After capturing this user's information, we retrieved data on all users in their following and fan lists. We then crawled these users' fan lists and the following lists. By July 10, 2017, we had captured data on 4,376,500 users.

After data cleaning—including tasks such as encoding, sorting, replacing missing values, and deleting duplicate values—we obtained 4,290,484 clean records. In addition to a name and profile picture, the user can provide a personal profile, which consists of a short text description that allows users to describe themselves. For example, the text could disclose their location, hobbies, personality traits, or physical appearance, or it could list their social accounts on other platforms. Next, we preprocessed the text data. Stop words were filtered out, and all symbols were removed except for Chinese characters, as we focused on single-language text. Finally, we performed a tokenization, lemmatization, and term-document matrix construction operation.

4.2.2. English long text corpus from Fiverr.com

We used a Python script to collect service descriptions from 7,491 users in the "lifestyle" category on Fiverr.com (Fig. 3). Services in this category include several subcategories, such as "spiritual & healing," "your message on," "online lessons," "traveling," "health," "nutrition &

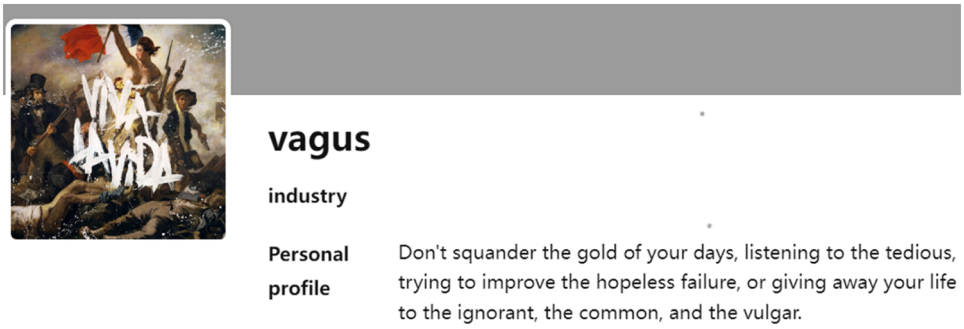


Fig. 2. User Interface on Zhihu.com.

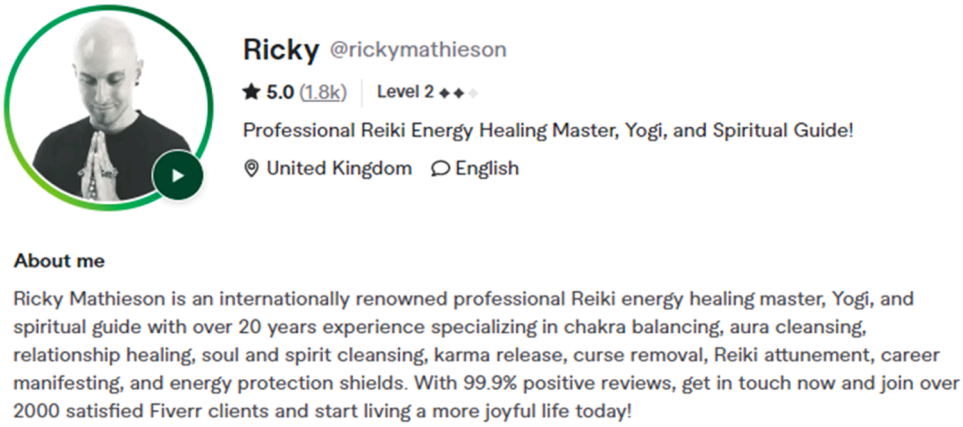


Fig. 3. User Interface on Fiverr.com.

fitness," "greeting cards & videos," "gaming," "family & genealogy," "collectibles," "celebrity impersonators," "astrology & psychics," "arts & crafts," "cooking lessons," "craft lessons," and "fitness lessons." After data cleaning, we finalized a corpus containing 7,491 entries. Next, we preprocessed the text data. We used Python's regular expressions to remove nontextual information and used the PyEnchant library[2] to spell-check the English text. We then used Natural Language Toolkit (NLTK)[3] for stemming and lemmatization. On this basis, we removed stop words, such as "a" and "to." Finally, we constructed a term-document matrix.

4.3. Results for VI-EM LDA

4.3.1. Metrics for selecting the optimal number of topics

When selecting the optimal number of topics for topic modeling, researchers commonly rely on two key metrics: perplexity and coherence. Perplexity measures how well a probability model predicts a sample, with lower values indicating better predictive performance, making it a statistical indicator of model quality:

$$\text{Perplexity}(W) = e^{-\frac{1}{N}\sum_{i=1}^{N}\log P(w_i)}, \tag{21}$$

where $\log P(w_i)$ is the logarithm of the probability of each word w_i, and the average of these log probabilities is taken over all the words in the test set.

However, perplexity alone is not always the best measure of model quality, especially in topic modeling, because it does not measure how meaningful or interpretable the topics are to humans. Hence, researchers often use it in conjunction with coherence scores. Coherence assesses the interpretability and meaningfulness of the topics generated by a model. A higher coherence score suggests that the words within each topic are more semantically related, making the topics themselves more understandable and relevant to human readers. Known as the document co-occurrence coherence measure, UMass relies on how frequently pairs of words appear together within the same documents. The score for each pair of words is computed by taking the log probability of their joint appearance, normalized by the probability of the individual occurrence of one of the words. The overall UMass score for a topic is the average of these scores for all word pairs, with values closer to zero (or more positive) indicating more coherent topics. This implies that the words are semantically related and appear together more often, enhancing topic interpretability. UMass is particularly effective for traditional topic modeling techniques, such as LDA, which depend heavily on word co-occurrence within documents. The calculation formula is as follows [50]:

$$C(k; V^{(t)}) = \sum_{m-2}^{M}\sum_{l=1}^{m-1}\log\frac{D\left(V_m^{(k)}, V_l^{(k)}\right)+1}{D\left(V_l^{(k)}\right)}. \tag{22}$$

This score represents the ratio of the frequency of one word and the frequency of another word contained in the words of all topics M under a

[2] PyEnchant is a python package used to check the spelling of words and suggest corrections for words that are misspelled. It can use many popular spellchecking packages to perform this task, including ispell, aspell, and MySpell. It is quite flexible at handling multiple dictionaries and multiple languages. The link is https://pypi.org/project/pyenchant.

[3] NLTK is a leading platform for building Python programs to work with human language data. It provides easy-to-use interfaces to over 50 corpora and lexical resources such as WordNet, along with a suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, semantic reasoning, and wrappers for industrial-strength NLP libraries. Here is the link for NLTK: https://www.nltk.org.

certain topic k. Then the logarithm of this ratio and the normalization operation are taken. Because human interpretation of topics often relies on word co-occurrence frequencies within the same topic, UMass provides valuable insights into topic interpretability. UMass is a negative value; the closer it is to zero, the better the topic has been learned.

Together, these metrics provide a balanced approach to determining the most appropriate number of topics by combining quantitative predictions with qualitative insights. As a result, we found the optimal number of LDA topics by calculating the UMass coherence score and the perplexity score.

4.3.2. *Chinese short text corpus from Zhihu.com*

We plotted the UMass score curve for the corpus from Zhihu.com, as shown in Fig. 4. The results indicate that the learning effect of the LDA was optimal when the number of topics is nine.

We also plotted the perplexity score curve for the Zhihu.com corpus. The result is shown in Fig. 5.

Although the perplexity score reached its lowest value of –8.6357 when the number of topics was 14, we found that when the number of topics was 9, the perplexity score was –8.0143. This score was the first value after a significant improvement in perplexity and was also lower than the perplexity score of –7.9521 for 8 topics. After that, the trend from 9 to 14 shows a gradual deceleration in the improvement of confusion, and the large number of topics may make it difficult to distinguish and interpret among topics. Based on these observations and the coherence results, we conclude that 9 is the optimal number of topics.

Next, we plotted the convergence curve. The VI-EM model reached convergence after 40 iterations, as shown in Fig. 6. We list the top 14 words contained in the 9 topics generated by the VI-EM LDA in Table 8.

4.3.3. *English long text corpus from Fiverr.com*

First, we plotted the UMass coherence score curve for the corpus from Fiverr.com to determine the optimal number of VI-EM LDA topics. The results, shown in Fig. 7, indicate that the learning effect of the LDA model is optimal when the number of topics is nine.

We also plotted the perplexity score curve for the Fiverr.com corpus, as shown in Fig. 8.

Results show that when the number of topics is nine, the perplexity score is lowest (–8.3401), indicating that the model fits the data well.

Next, we plotted the convergence curve for the English long-text corpus in Fig. 9. The VI-EM model reached convergence after 48 iterations. We then listed the top 14 words contained in the 9 topics generated by the VI-EM LDA in Table 9.

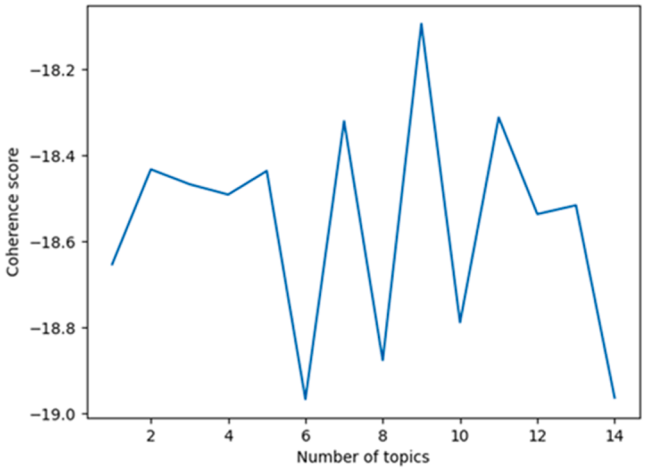


Fig. 4. Relationship between UMass score and number of topics (Zhihu.com).

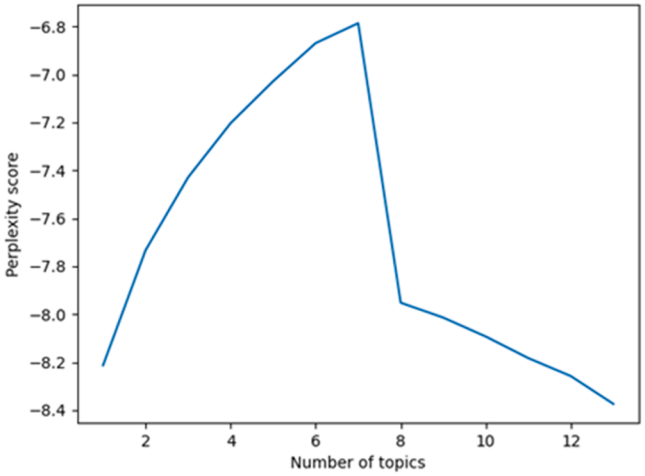


Fig. 5. Relationship between perplexity score and number of topics (Zhihu.com).

4.4. *Result for the GMM*

We conducted the GMM using vectors vectorized by Word2vec. To determine the optimal number of clusters (number of components), we used the Akaike information criterion (AIC) to choose a model that fits the data well without using too many components. It balances model fit with complexity, by calculating each model's score based on the likelihood of the model (how well it fits the data) and the number of parameters it uses (the complexity of the model), as defined by the formula:

$$\mathrm{AIC} = 2k - 2\ln(L), \tag{23}$$

where k is the number of parameters in the model, and $\ln(L)$ is the log-likelihood of the model. Lower AIC scores indicate a better fit without overfitting.

4.4.1. *Chinese short text corpus from Zhihu.com*

For the Zhihu.com corpus, we calculated the AIC for different numbers of components in the GMM and plotted the result in Fig. 10. To determine the optimal number of topics for the LDA-GMM-CorEx model, we employed model selection metrics, such as the elbow method on log-likelihood plots and topic coherence scores, to identify a range of plausible topic numbers. We then selected the final number of topics based on two similar choices.

The AIC results showed that four and seven are the optimal clustering numbers, however, we chose to proceed with seven clusters over four for several reasons. While both four and seven clusters might fit the data well according to AIC, seven clusters might cover more variability within the dataset. This can be particularly important in multidimensional data where many variables influence the structure. More clusters can capture these complexities more effectively, providing a more comprehensive understanding of the interactions and relationships within the data. Additionally, selecting seven clusters allows for a more granular breakdown of the data. This finer segmentation can reveal more specific and nuanced insights that might be obscured with only four clusters. The additional clusters provide a deeper differentiation between subgroups or patterns within the dataset, which can be particularly valuable in complex or diverse datasets where the distinctions between data points are subtle. This level of detail is particularly valuable in complex datasets, where subtle thematic variations are significant. The clustering results are presented in Table 10.

Upon further qualitative analysis of the clusters, we observed a significant overlap in the topic matter between clusters 1 and 6 based on a detailed semantic analysis of the top contributing words and their

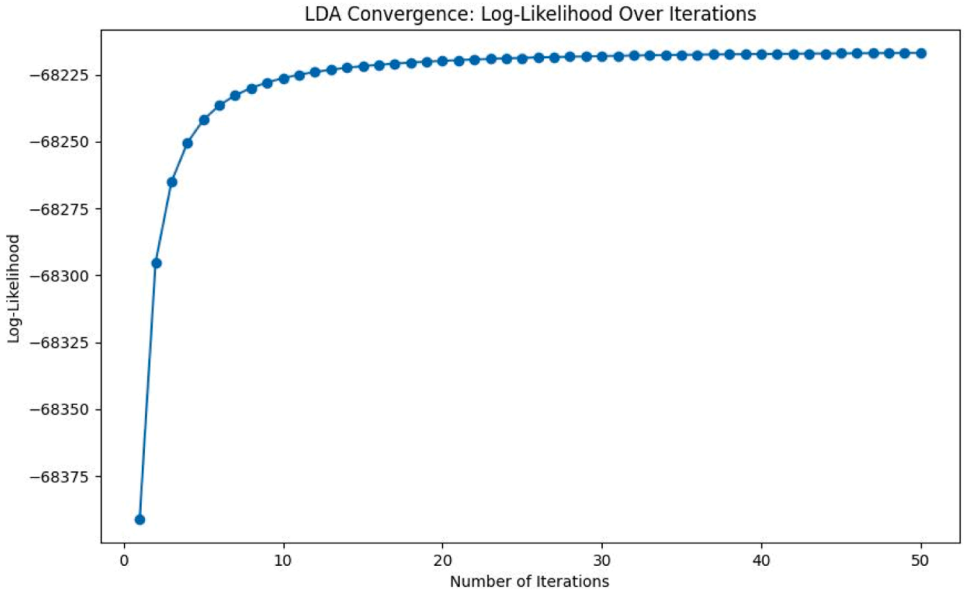


Fig. 6. Relationship between log-likelihood and number of iterations (Zhihu.com).

Table 8
Results for VI-EM LDA Topic Model (Zhihu.com).

Topic	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8	Topic 9
Word 1	公众 (Public)	爱好者 (Hobbyist)	工程师 (Engineer)	学生 (Student)	金融 (Finance)	微信 (WeChat)	大学生 (College student)	专业 (Professional)	摄影 (Photography)
Word 2	管理 (Administration)	在读 (Studying)	大学 (University)	老师 (Teacher)	互联网 (Internet)	关注 (Following)	生活 (Life)	学习 (Study)	努力 (Effort)
Word 3	医学生 (Medical student)	设计师 (Designers)	会计 (Accounting)	博士 (PhD)	教师 (Teacher)	技术 (Technology)	音乐 (Music)	健身 (Fitness)	世界 (World)
Word 4	程序 (Program)	喜欢 (Like)	游戏 (Game)	少女 (Girl)	研究生 (Graduate)	医学 (Medical)	研究 (Research)	医生 (Doctor)	电影 (Film)
Word 5	教育 (Education)	设计 (Design)	自由 (Free)	故事 (Story)	程序员 (Programmer)	知识 (Knowledge)	患者 (Patient)	青年 (Youth)	人生 (Life)
Word 6	科技 (Technology)	心理学 (Psychology)	旅游 (Travel)	在读 (Reading)	爱好 (Hobby)	信号 (Signal)	本科 (Undergraduate)	文艺 (Literature and art)	高中生 (Senior high school student)
Word 7	咨询 (Consulting)	汽车 (Automotive)	产品 (Product)	计算机 (Computer)	媒体 (Media)	专注 (Focus on)	历史 (History)	高三 (Senior year)	法学 (Law)
Word 8	行业 (Industry)	文学 (Literature)	运营 (Operation)	专业 (Major)	业余 (Amateur)	故事 (Story)	销售 (Sales)	资深 (Senior)	读书 (Reading)
Word 9	留学 (Study abroad)	本科 (Undergraduate)	旅行 (Travel)	梦想 (Dream)	职业 (Occupation)	美食 (Food)	律师 (Lawyer)	创始人 (Founder)	英语 (English)
Word 10	旅游 (Tourism)	旅游 (Tourism)	工科 (Engineering)	高三 (Senior year)	数学 (Mathematics)	心理学 (Psychology)	留学 (Study abroad)	关注 (Following)	有趣 (Interesting)
Word 11	未来 (Future)	摄影 (Photography)	经理 (Manager)	研究 (Research)	工程 (Engineering)	媒体 (Media)	梦想 (Dream)	旅行 (Travel)	未来 (Future)
Word 12	投资 (Investment)	博士 (PhD)	专注 (Focusing on)	大学 (University)	民工 (Migrant worker)	投资 (Investment)	艺术 (Art)	旅游 (Travel)	小白 (Beginner)
Word 13	工程 (Engineering)	法学 (Law)	建筑 (Construction)	法学 (Law)	从业者 (Practitioner)	生活 (Life)	计算机 (Computer)	单身 (Single)	热爱 (Love)
Word 14	文学 (Literature)	游戏 (Games)	美食 (Delicious food)	艺术 (Art)	法律 (Law)	学习 (Learning)	考研 (Graduate school exam)	留学 (Study abroad)	创业 (Entrepreneurship)

contextual relevance within the dataset. This overlap can make separate clusters redundant, leading to similar or duplicate topics. Therefore, we merged topic 1 and topic 6 to reduce redundancy and strengthen the overall interpretability of our model. Then, we selected the top three words in each topic as anchor words and summarized each topic. The anchor words are presented in Table 11:

4.4.2. *English long text corpus from Fiverr.com*

We conducted the GMM using vectors vectorized by Word2vec for the Fiverr.com corpus. First, we used the AIC method to determine the optimal number of clusters, as shown in Fig. 11.

The AIC results showed that 8 and 9 were the optimal clustering numbers. By applying the criteria used previously for selecting the optimal cluster number in the Chinese short text corpus, we chose 9 as our optimal cluster number to maximize the granularity and precision of

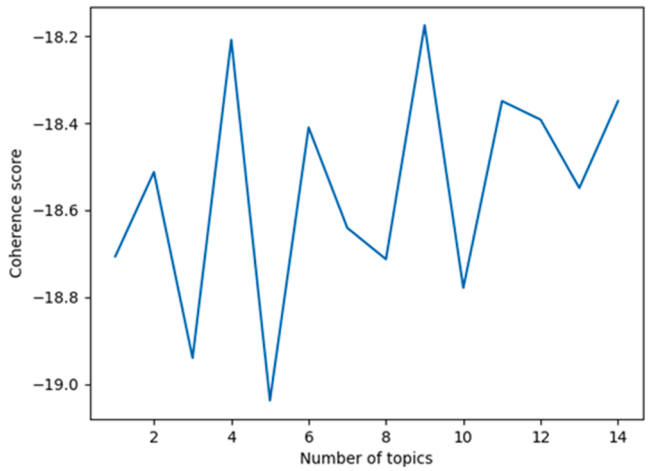


Fig. 7. Relationship between UMass score and number of topics (Fiverr.com).

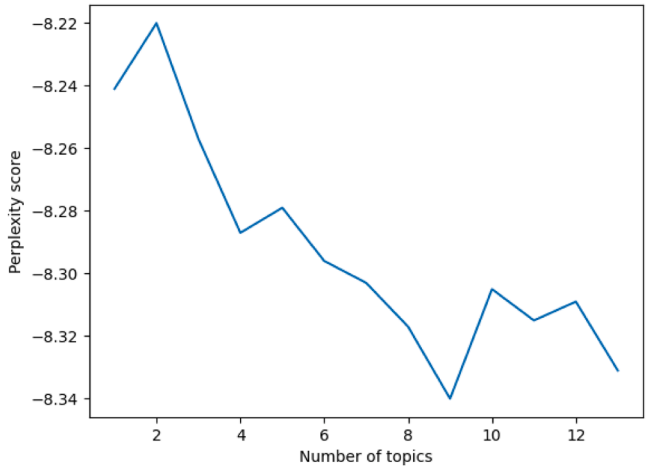


Fig. 8. Relationship between perplexity score and number of topics (Fiverr.com).

our analysis. The clustering results are shown in Table 12:

To enhance the results, we removed duplicate words and the topic of non-named entity words (e.g., Topic 6). Finally, we selected the top three words in each topic as anchor words, which are summarized in Table 13:

4.5. *Result for the CorEx model*

4.5.1. *Chinese short text corpus from Zhihu.com*

We input the anchor words from Table 13 into the CorEx topic model according to their corresponding categories. The model reached convergence after 150 iterations, as shown in Fig. 12.

We output the final topic modeling results in Table 14, which contains the top five topic words and the multivariate mutual information values for each topic. These words represent those with the highest multivariate mutual information within their respective topic, rather than the highest within-topic probabilities, as in LDA. For example, the topic words "design," "engineering," "financial," "programming," and "Internet" effectively summarize Topic 1: Industry. We found that the LDA-GMM-CorEx topic model generated the corresponding topics according to the seven predefined categories. Moreover, there is no repetition of words between topics, which solves the problem of model interpretability and topic coincidence.

4.5.2. *English long text corpus from Fiverr.com*

We then input the anchor words shown in Table 11 into the CorEx topic model. After 150 iterations, the model reached convergence, as shown in Fig. 13.

Results of the LDA-GMM-CorEx topic model are shown in Table 15, which contains the top five topic words and the multivariate mutual information values for each topic. For example, the topic words "spell," "healing," "spiritual," "psychic," and "powerful" effectively summarize Topic 1: Psychic.

The results demonstrate that the topics generated by our model are closely related to the categories (Psychic, Game, Message, Photo, and Health) and aspects (Time, Order, and Question) of gig services. This validates the model's ability to embed domain knowledge and generate low-redundancy, highly interpretable topics.

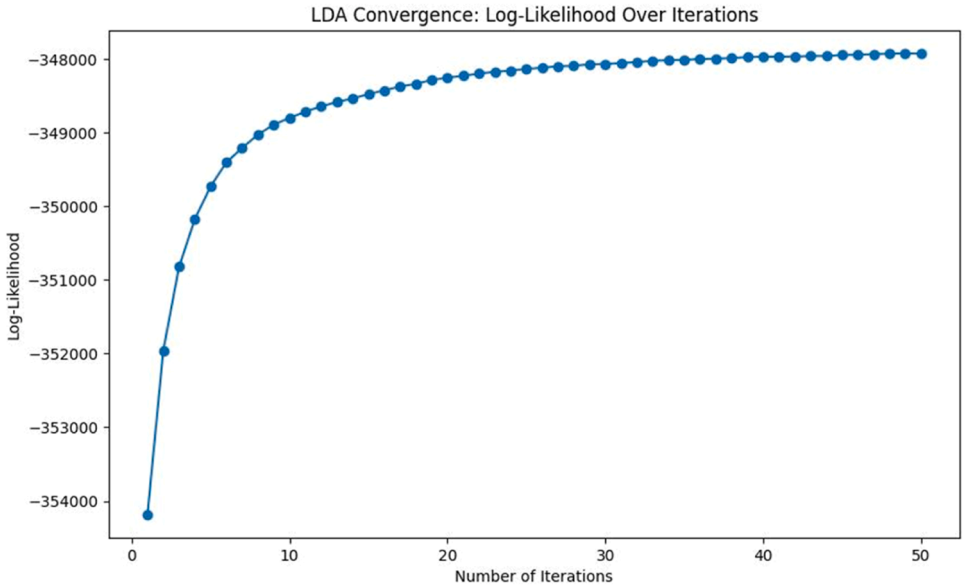


Fig. 9. Relationship between log-likelihood and number of iterations (Fiverr.com).

Table 9
Result for VI-EM LDA topic model (Fiverr.com).

Topic	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8	Topic 9
Word 1	Reading	Recipe	Help	Question	Video	Spell	Life	Energy	Video
Word 2	Help	Help	Plan	Logo	Help	Love	Reading	Healing	Game
Word 3	Give	Question	Gig	Video	Professional	Gig	Help	Reiki	Play
Word 4	Time	Image	Photo	Best	Need	Life	Love	Session	Birthday
Word 5	Like	Reading	Text	Year	Order	Help	Need	Body	Help
Word 6	Know	Like	Diet	Card	Question	Need	Question	Help	Message
Word 7	Problem	Love	Life	Work	Health	Cast	New	Feel	Player
Word 8	Life	Give	Year	Experience	Work	Just	Relationship	Negative	Gig
Word 9	Need	Answer	Need	Help	Answer	Powerful	Year	Spiritual	Picture
Word 10	Gig	Know	Work	Message	Here	Know	Time	Life	Happy
Word 11	Card	Gig	Time	Order	Contact	Back	Feel	People	Need
Word 12	Relationship	Time	Wish	Design	Year	Magic	Experience	Work	Only
Word 13	Work	Tell	Order	Give	Name	People	People	Spirit	Word
Word 14	Message	Work	Don't	Contact	Reading	Work	Psychic	Aura	Give

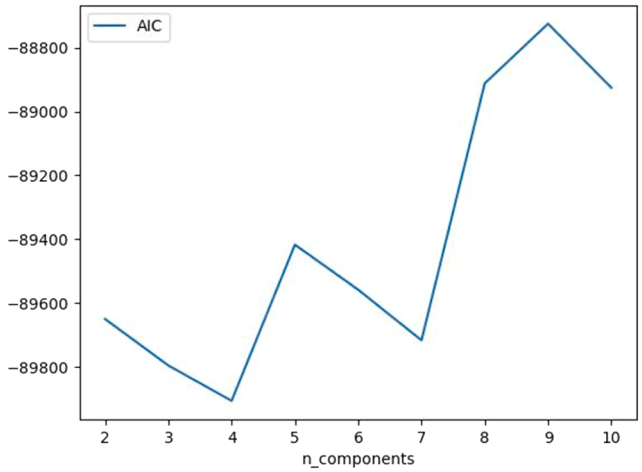


Fig. 10. Relationship between AIC value and the number of components (Zhihu.com).

4.6. Evaluation results

4.6.1. Evaluation metrics and benchmark model

Because the CorEx topic model is not generative but judgmental, traditional measures such as perplexity do not apply to the topic modeling approach in this paper. Instead, the coherence eigenvalue is a better proxy for the comprehensibility of topic headings. Although perplexity measures the model's predictive performance, it does not

Table 11
Anchor words generated by LDA-GMM (Zhihu.com).

	Anchor Word 1	Anchor Word 2	Anchor Word 3
Cluster 1	设计师 (Designer)	工程师 (Practitioner)	经理 (Manager)
Cluster 2	学生 (Student)	大学生 (College student)	高中生 (High school student)
Cluster 3	文学 (Literature)	心理学 (Psychology)	法学 (Law)
Cluster 4	游戏 (Game)	旅游 (Travel)	健身 (Fitness)
Cluster 5	努力 (Try hard)	梦想 (Dream)	自由 (Liberty)
Cluster 6	微信 (WeChat)	公众号 (Public account)	关注 (Following)

Table 10
Result for GMM (Zhihu.com).

Topic	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7
Word 1	学生(Students)	文学 (Literature)	设计师 (Designer)	自由 (Freedom)	游戏 (Games)	本科 (Undergraduate)	微信 (WeChat)
Word 2	大学生 (College student)	心理学 (Psychology)	工程师 (Engineer)	梦想 (Dream)	音乐 (Music)	博士 (PhD)	公众号 (Public number)
Word 3	高中生 (Senior high school student)	法学 (Law)	经理 (Manager)	努力 (Try hard)	旅游 (Travel)	工科 (Engineering)	关注 (Following)
Word 4	高三 (Senior year)	建筑 (Architecture)	教师 (Teacher)	人生 (Life)	电影 (Movies)	在读 (In school)	信号 (Signal)
Word 5		艺术 (Art)	程序员 (Programmer)	未来 (Future)	美食 (Food)	研究生 (Graduate student)	媒体 (Media)
Word 6		数学 (Mathematics)	从业者 (Practitioner)		健身 (Fitness)		运营 (Operation)
Word 7		法律 (Law)	律师 (Lawyer)		文艺 (Literature)		互联网 (Internet)
Word 8		技术 (Technology)	医生 (Doctor)		旅行 (Travel)		
Word 9		医学 (Medical)	创始人 (Founder)		摄影 (Photography)		
Word 10		历史 (History)			读书 (Reading)		
Word 11					小白 (Beginner)		
Word 12					热爱 (Love)		

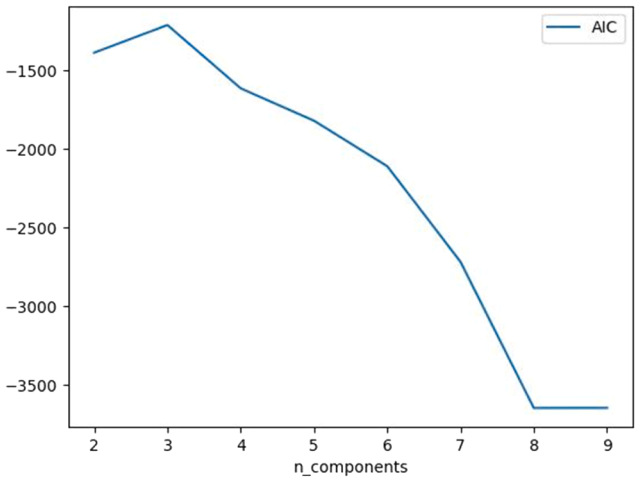


Fig. 11. Relationship between AIC value and number of components (Fiverr.com).

Table 12
Results for GMM (Fiverr.com).

Topic	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8	Topic 9
Word 1	Spell	Day	Game	Message	Professional	Feel	Answer	Life	Photo
Word 2	Psychic	Time	Like	Text	Order	Give	Problem	Health	Picture
Word 3	Healing	Birthday	Play	Message	Contact	Help	Question	Energy	Logo
Word 4	Reiki	Year	Happy			Need			
Word 5	Spiritual	Session	Love			Know			

Table 13
Anchor words generated by LDA-GMM (Fiverr.com).

	Anchor Word 1	Anchor Word 2	Anchor Word 3
Cluster 1	Spell	Psychic	Healing
Cluster 2	Day	Time	Year
Cluster 3	Professional	Order	Contact
Cluster 4	Game	Like	Play
Cluster 5	Message	Text	Message
Cluster 6	Photo	Picture	Logo
Cluster 7	Question	Answer	Ask
Cluster 8	Health	Life	Energy

reflect how interpretable the topics are to humans [51]. This is primarily because perplexity measures how well the model predicts unseen documents but does not consider whether the topics make sense to users. Topics with lower perplexity can still be semantically disjointed or lack meaningful interpretation. Studies have shown that there is often a lack of correlation between perplexity and human judgment of topic quality. For instance, Chang et al. [52] found that models with better perplexity did not always align with human evaluations of topic coherence and interpretability. This suggests that optimizing solely for perplexity might lead to statistically probable topics that are not useful for understanding the underlying themes of the documents.

Coherence, on the other hand, specifically measures the semantic similarity between high-scoring words within the same topic, providing a direct indication of what humans might consider a "good" topic. [51]. Empirical evidence shows that higher topic coherence corresponds to more interpretable topics, as judged by human evaluators [50]. Therefore, we employed topical coherence scores to compute semantic coherence. Considering that our model differs from the traditional LDA

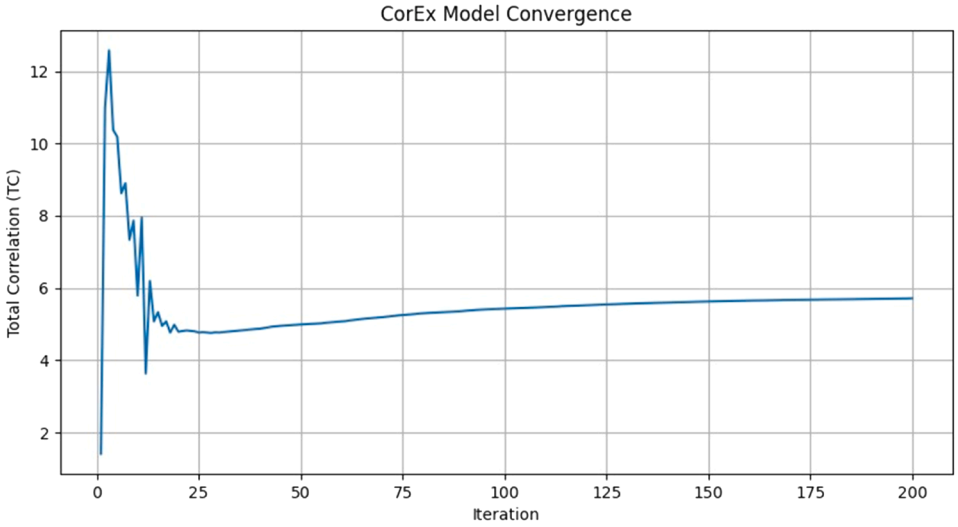


Fig. 12. Iteration and convergence (Zhihu.com).

Table 14
Results for the CorEx model (Zhihu.com).

Topic	Word 1	Word 2	Word 3	Word 4	Word 5
Topic 1: Account	公众号	微信	关注	运营	互联网
	(Public account)	(WeChat)	(Following)	(Operation)	(Internet)
	0.359	0.212	0.095	0.078	0.028
Topic 2: Industry	工程师	设计师	经理	律师	软件设计师
	(Engineer)	(Designer)	(Manager)	(Lawyer)	(Software designer)
	0.282	0.233	0.079	0.056	0.008
Topic 3: Education	学生	大学生	博士	高中生	小学生
	(Student)	(College student)	(Phd)	(High school student)	(Elementary school student)
	1.510	0.169	0.075	0.052	0.006
Topic 4: Major	医学	心理学	法学	文学	理学
	(Medical)	(Psychology)	(Law)	(Literature)	(Science)
	0.158	0.080	0.073	0.066	0.019
Topic 5: Hobby	爱好	喜欢	摄影	健身	电影
	(Hobbies)	(Like)	(Photography)	(Fitness)	(Movies)
	0.579	0.200	0.155	0.124	0.004
Topic 6: Sense of Value	自由	努力	人生	梦想	不羁
	(Freedom)	(Work hard)	(Life)	(Dream)	(Uninhibited)
	0.146	0.129	0.110	0.062	0.003

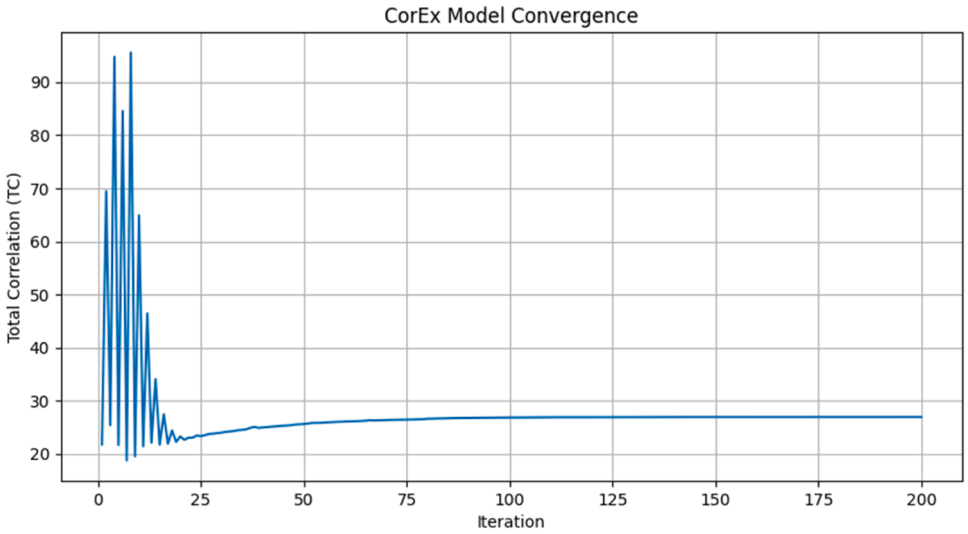


Fig. 13. Iteration and convergence (Fiverr.com).

Table 15
Results for CorEx model (Fiverr.com).

Topic	Word 1	Word 2	Word 3	Word 4	Word 5
Topic 1: Psychic	Spell	Healing	Spiritual	Psychic	Powerful
	0.8235	0.4834	0.3934	0.3449	0.0723
Topic 2: Time	Time	Year	Day	Session	Experience
	1.0907	0.9194	0.582	0.2748	0.0361
Topic 3: Order	Order	Contact	Professional	Service	Ordering
	0.9281	0.8004	0.4807	0.4495	0.0079
Topic 4: Game	Like	Game	Play	PC	Gaming
	1.4791	0.47	0.3877	0.0067	0.0056
Topic 5: Message	Message	Text	Drop	Send	Innovative
	1.465	0.5357	0.003	0.0027	0.0025
Topic 6: Photo	Photo	Picture	Logo	Card	Image
	0.6438	0.5015	0.4879	0.3961	0.0357
Topic 7: Question	Question	Ask	Answer	Reading	Accurate
	1.4835	0.6872	0.6526	0.0314	0.0091
Topic 8: Health	Life	Energy	Health	Positive	Wealth
	1.5354	0.7967	0.4311	0.0254	0.0086

model, and the CorEx model still lacks a mature Python package for consistency calculation, we developed a customized implementation to calculate consistency metrics, ensuring compatibility with our model's unique framework and providing more accurate and reliable evaluations.

In this paper, we employ the UMass and C_V coherence scores to evaluate our model's performance. We specifically selected the UMass and C_V metrics for evaluating the quality of topic models because they minimize the influence of the number of topics, thereby ensuring fair and consistent evaluations. Specifically, both metrics assess a fixed number of top words (e.g., the top-10 words) from each topic, which allows us to focus on the quality and coherence of individual topics rather than the overall number of topics in the model. This methodology ensures that our evaluation is independent of the total number of topics generated. Moreover, these metrics do not directly rely on the number of topics produced by the model but instead evaluate the internal consistency and coherence of each topic. They also effectively manage sparsity in co-occurrence matrices or embeddings, ensuring reliable coherence calculations across different numbers of topics in various models.

These scores are crucial for calculating the coherence of topics, each serving distinct purposes and offering unique insights. UMass helps validate the internal consistency of topics, aiding in the assessment of clarity and usability. However, it primarily captures document-level relationships and may not fully capture the complexity and nuances of the topics generated, especially those generated from datasets with sparse co-occurrence or short documents. In contrast, the C_V score provides a more comprehensive evaluation by considering both the

frequency and similarity of words across the entire text corpus, not just within individual documents. This score is derived from a sliding window technique that captures broader contextual relationships, making it well suited to models that encode deeper linguistic patterns. The C_V score is invaluable for assessing topics from models with richer contextual embeddings, offering a broader view of topic quality beyond mere co-occurrence. The coherence score is calculated as follows:

$$C_V = \sum_{m-2}^{M} \sum_{l=1}^{m-1} \frac{\sum_{i=1}^{K} \mathrm{NPMI}(\omega_m, \omega_l) \cdot \mathrm{cosinesimilarity}(\omega_m, \omega_l)}{M \cdot (M-1)/2}. \quad (24)$$

NPMI is the normalized pointwise mutual information between words ω_m and ω_l. M is the number of top words considered for the topic, and K is the number of documents in which both ω_m and ω_l appear.

By integrating the UMass and C_V scores, we aim to harness their respective strengths: UMass's focus on document-level coherence and C_V's broader contextual analysis. This dual approach ensures a thorough and nuanced evaluation of our topic models, accommodating both traditional and more complex structures.

To further validate these findings and explore how these metrics perform under different modeling paradigms, we extended our analysis to include a broader comparison of topic modeling techniques. In addition to single LDA, LDA combined with the GMM, and CorEx, we also compared the LDA-GMM-CorEx topic model with the large language model BERT. BERT's bidirectional training architecture allows it to integrate context from both the left and right sides of a token within any layer, which is pivotal for comprehensively understanding the nuances in textual data [53]. This characteristic makes BERT particularly suitable for applications involving nuanced topic extraction from complex documents, outperforming other models in tasks involving specific or subtle contextual interpretation [54].

Moreover, BERT's design for predicting masked words in a text inherently aligns with tasks requiring deep semantic understanding, such as topic modeling [55]. Studies have demonstrated BERT's effectiveness in enhancing topic coherence and distinctiveness, which are critical for the precise extraction and categorization of topics within various texts, making it a more relevant benchmark for evaluating our hybrid model's performance in topic modeling tasks [56]. Topic modeling with BERT uses a pretrained BERT model to extract feature representations of text, known as text embeddings, and then uses the extracted embeddings for K-mean clustering to discover hidden topics in the document set.

4.6.2. *Chinese short text corpus from Zhihu.com*

To further assess and summarize the models' performance, we analyzed the UMass and C_V scores for the Zhihu.com corpus, as presented in Table 16. Considering the importance of topic comparability for interpreting results more holistically, we operated all models on the same dataset and were evaluated under identical pre-processing and coherence scoring conditions. This consistency helps ensure that any differences in coherence scores are attributable to the models' performance rather than discrepancies in thematic alignment. To further enhance comparability, we calculated the average coherence score per topic for each model. Averaging the scores provides an overall measure of topic quality for each model, focusing on its ability to produce coherent topics. This approach avoids the subjectivity involved in topic alignment and ensures an objective comparison of models based on their coherence metrics.

We found that the UMass coherence score of the LDA-GMM-CorEx topic model was the closest to zero, and the C_V score of the model reached the highest value. This indicates that the topics generated by our proposed topic model were more meaningful and semantically coherent and had better aspect interpretability than other models. The coherence score results of the combined models of LDA and the GMM showed better coherence than those of the LDA model. This shows that the integration of LDA and the GMM is a powerful method for understanding complex text corpora, in which LDA's nuanced thematic analysis is leveraged by the GMM's clustering capabilities to reveal patterns and groupings that might not be immediately apparent. The combination of LDA with the GMM bridges the gap between topic modeling and document clustering, offering a comprehensive approach to text analysis.

An improvement in the UMass coherence score from –5.501 to –4.973 and in the C_V coherence score from 0.362 to 0.610 indicated a positive change in the coherence of the topics generated by our topic modeling algorithm. This means that, after optimization, the words within the topics were more meaningfully related to each other according to their co-occurrence in the same documents within the corpus.

We conducted a comparative analysis of coherence scores between our proposed LDA-GMM-CorEx topic model and the large language model BERT. The results indicated that the LDA-GMM-CorEx model outperformed the BERT-based model in terms of coherence. Specifically, the LDA-GMM-CorEx model improved the UMass coherence score from –6.479 to –4.973 and the C_V coherence score from 0.313 to 0.610, highlighting its effectiveness in generating more coherent and thematically consistent topics. These findings emphasize the advantages of

Table 17
Comparative analysis results of TC values (Zhihu.com).

Model	TC Value	
Topic	CorEx	VI-EM LDA + GMM + CorEx
Total	2.089	8.112
Topic 1	0.436	1.274
Topic 2	0.384	1.076
Topic 3	0.356	2.357
Topic 4	0.299	0.754
Topic 5	0.277	1.648
Topic 6	0.187	0.590
Topic 7	0.151	0.415

Table 16
Comparative analysis results of UMass score (Zhihu.com).

Model	Coherence							
	VI-EM LDA		VI-EM LDA + GMM		VI-EM LDA + GMM + CorEx		BERT	
Topic	UMass	C_V	UMass	C_V	UMass	C_V	UMass	C_V
Average	–5.501	0.362	–5.464	0.414	–4.973	0.610	–6.479	0.313
Topic 1	–6.282	0.341	–6.560	0.346	–3.432	0.787	–5.054	0.494
Topic 2	–4.827	0.513	–5.198	0.48	–6.492	0.428	–7.483	0.110
Topic 3	–5.690	0.275	–6.324	0.163	–6.461	0.561	–4.666	0.581
Topic 4	–5.396	0.309	–4.851	0.464	–5.254	0.588	–6.070	0.158
Topic 5	–5.526	0.352	–4.717	0.559	–3.412	0.761	–6.332	0.306
Topic 6	–5.274	0.400	–5.133	0.471	–4.789	0.537	–9.267	0.229
Topic 7	–5.617	0.281						
Topic 8	–5.598	0.386						
Topic 9	–5.295	0.396						

Table 18
Comparative analysis results of coherence score (Fiverr.com).

Model	Coherence							
	VI-EM LDA		VI-EM LDA + GMM		VI-EM LDA + GMM + CorEx		BERT	
Topic	UMass	C_V	UMass	C_V	UMass	C_V	UMass	C_V
Average	–2.028	0.444	–2.013	0.697	–1.948	0.712	–3.355	0.520
Topic 1	–2.207	0.533	–2.386	0.628	–2.107	0.648	–1.801	0.505
Topic 2	–1.888	0.432	–2.476	0.603	–2.610	0.454	–2.660	0.528
Topic 3	–2.000	0.366	–1.904	0.629	–2.801	0.566	–11.501	0.323
Topic 4	–2.049	0.429	–2.635	0.456	–2.880	0.734	–2.551	0.498
Topic 5	–1.888	0.455	–2.030	0.841	–1.066	0.873	–2.0700	0.524
Topic 6	–1.825	0.520	–1.994	0.819	–1.976	0.775	–1.377	0.624
Topic 7	–2.043	0.448	–1.330	0.890	–1.124	0.906	–2.291	0.545
Topic 8	–2.330	0.380	–1.351	0.712	–1.019	0.738	–2.588	0.613
Topic 9	–2.022	0.431						

Table 19
Comparative analysis results of TC values (Fiverr.com).

Model	TC Value	
Topic	CorEx	VI-EM LDA + GMM + CorEx
Total	14.0072	21.5922
Topic 1	3.1821	3.7360
Topic 2	2.2195	2.5585
Topic 3	1.9053	1.8968
Topic 4	1.7809	2.5749
Topic 5	1.4022	2.7201
Topic 6	1.2871	2.4076
Topic 7	1.1246	2.7539
Topic 8	1.1055	2.9443

using models specifically designed for topic modeling, because they are better equipped to capture the nuanced relationships within text data.

In addition, we compared the TC value between the CorEx model and the LDA-GMM-CorEx model. Higher TC values indicate that the CorEx topic model has been better trained, as they signify the creation of more informative topics related to the document. The TC values are presented in Table 17:

The total correlation value of the CorEx topic model was 2.089, whereas the TC value of the LDA-GMM-CorEx topic model was 8.112, indicating that the LDA-GMM-CorEx topic model generated more topics about document information and outperformed the CorEx model.

4.6.3. English long text corpus from Fiverr.com

We compared the UMass scores of the VI-EM LDA; the hybrid model of VI-EM LDA and the GMM; the hybrid model of VI-EM LDA, the GMM, and CorEx; and the BERT model to prove the validity of our model. The UMass and C_V scores are summarized in Table 18:

We found that the UMass coherence score of the LDA-GMM-CorEx topic model was the closest to zero, and the C_V score was the highest, indicating that it generated more meaningful and semantically coherent topics with better interpretability. An improvement in the UMass coherence score from –2.028 to –1.948 and in the C_V coherence score from 0.444 to 0.712 demonstrates a positive change in topic coherence. Furthermore, the LDA-GMM-CorEx model outperformed the BERT-based model, with a UMass coherence score improvement from –3.355 to –1.948 and a C_V coherence score improvement from 0.520 to 0.712, further demonstrating the validity of our hybrid model.

In addition, we compared the TC value between the CorEx model and the LDA-GMM-CorEx model. The TC values are summarized in Table 19:

The TC value of the CorEx topic model was 14.0072, whereas that of the LDA-GMM-CorEx topic model was 21.5922. The result shows that our new topic model was superior to the CorEx topic model.

5. Discussion

We discovered that the LDA model effectively generates vocabularies related to prior domain knowledge [18]. The operation works fast both on a large dataset containing 4,290,484 short-text personal profiles and on a dataset containing 7,291 long-text service-related introductions. More importantly, these generated vocabularies provided a quick overview of the aspects contained in the text, enabling us to swiftly organize desired topic categories and the corresponding vocabulary [52]. We found that the topics generated by the LDA-GMM-CorEx model effectively explained the areas covered by the corpus. Specifically, the model generated six personal information topics based on the Zhihu.com dataset: Account, Industry, Education, Major, Hobby, and Sense of Value. Based on the Fiverr.com dataset, we generated eight service-related topics: Psychic, Time, Order, Game, Message, Photo, Question, and Health.

Our prior domain knowledge topic modeling also passed the validity test. First, in both cases, the TC of the LDA-GMM-CorEx topic model was greater than that of the CorEx topic model. This suggests that anchor words created by combining the topic words generated by LDA with researchers' domain knowledge performed better than anchor words selected solely based on their highest multivariate mutual information with the label. The former generated more topics about document information [43].

Additionally, we compared the coherence scores between the VI-EM LDA model, the hybrid model of LDA and the GMM, the large language model BERT, and the LDA-GMM-CorEx topic model. Our findings indicate that our proposed model had the best performance on interpretability and semantic coherence, aligning with recent findings on the importance of model coherence in topic modeling [51]. We also compared the performance of the LDA-GMM-CorEx model with that of the BERT neural embedding model, a leading topic modeling approach, particularly in handling complex data sets. Our comparative analysis revealed that the LDA-GMM-CorEx model demonstrated superior performance in terms of coherence scores, aligning with recent findings on the importance of model coherence in topic modeling [57]. Specifically, the model achieved higher UMass and CV coherence scores than BERT [19], highlighting its effectiveness in capturing the semantic relationships necessary for high-quality topic modeling. Although BERT and similar large language models are proficient at capturing deep contextual relationships within text because of their transformer architecture [53], they require substantial computational resources. They are not inherently structured to generate discrete topic classifications without substantial customization [58]. In contrast, our LDA-GMM-CorEx model categorizes text efficiently into clearly interpretable topics, making it particularly useful for applications requiring quick and clear topic extraction and categorization. This efficiency and clarity in topic generation make our model a preferable option for practical implementations where computational resources and immediate applicability are key considerations [53].

Finally, we compared the performance differences of the models for the two datasets. In terms of TC value and coherence score, the LDA-GMM-CorEx model performed better on English long-texts than on Chinese short-texts. This discrepancy likely arises because LDA was originally designed for longer documents and CorEx has been optimized for English, impacting its effectiveness on non-English datasets.

However, our model showed significant improvement in the Chinese short-text corpus over the English long-text corpus. We calculated the growth rates of the TC value and the UMASS score for the LDA-GMM-CorEx model in both corpora—that is, the growth amount divided by the original value. Compared to the VI-EM LDA model, the UMASS score of the LDA-GMM-CorEx model decreased by 0.096, and the TC value increased by 2.883 in the Chinese short-text corpus. The UMass score of the LDA-GMM-CorEx model for the English long-text corpus decreased by 0.040, and the TC value increased by 0.542. These results show that the LDA-GMM-CorEx model was better at improving the performance of nonstandard data sets (e.g., Chinese short-texts).

The LDA-GMM-CorEx efficiently extracts domain-knowledge-driven categories, compensating for the limitations of traditional topic modeling methods. It alleviates the problem of poor interpretability, such as topic overlap and difficulty summarizing, caused by traditional topic modeling methods. Moreover, this method automatically generates anchor words that cover domain knowledge, compensating for the shortcomings of the CorEx model. Most importantly, this domain-knowledge-driven approach improves vocabulary distribution characteristics without compromising topic quality [13].

5.1. Implications for theory and practice

The LDA-GMM-CorEx model offers distinct theoretical implications for IS and management. First,the hybrid model theoretically advances the integration of probabilistic, statistical, and information-theoretic methods to improve topic modeling. LDA primarily utilizes a probabilistic approach for topic modeling, which may not handle continuous data efficiently or capture complex data dependencies [18]. Our LDA-GMM-CorEx hybrid model extends beyond these limitations by integrating the GMM and CorEx. The GMM improves the model's capability to manage continuous data [59], whereas CorEx adds an information-theoretic layer that aids in uncovering more nuanced data relationships [13]. This integration not only broadens the theoretical foundations of topic modeling but also leads to a more robust framework for understanding how different data characteristics influence topic discovery and categorization in IS.

Second, the theoretical contributions of the LDA-GMM-CorEx model lie in its innovative integration of dynamic adaptability mechanisms, which significantly extend the functional scope of traditional topic models in information systems. Traditional static models have limitations in adapting to evolving data [60], which require models that can adjust to new information dynamically. This is particularly important in fields with rapid data evolution, such as IS [13,18]. To address this, the LDA-GMM-CorEx model enhances adaptability by including the GMM, which allows the model to respond dynamically to new topics and vocabulary shifts [60]. This capability theoretically supports the development of more resilient IS models that maintain their effectiveness despite underlying data variability. It contributes to theories that explain how information systems can sustain performance and accuracy in dynamic environments, such as e-commerce, where data variability is prominent because of fluctuating user preferences and market conditions.

Third, our hybrid model improves interpretability through domain-specific integration. Traditional topic modeling tools can yield results that are difficult to interpret, limiting their usefulness for decision-making in IS [52]. By incorporating CorEx, our model boosts interpretability significantly, making the insights it generates more relevant and applicable for decision support systems [43]. This improvement provides a foundation for developing theories on how enhanced interpretability of data-driven models can lead to better-aligned and more actionable insights in organizational settings, where such insights are crucial for strategic and operational decision-making.

Finally, resolution of semantic ambiguities enhances our understanding of textual data complexity. Standard topic modeling approaches may not distinguish adequately between semantically similar terms, which can lead to ambiguities in understanding textual data [61]. Incorporating the GMM enhances the model's ability to make finer distinctions between semantically similar terms, addressing the common linguistic challenges of polysemy and synonymy [62]. Theoretically, this enhancement contributes to a deeper understanding of semantic processing within IS and offers insights into how models can be designed to improve the precision of meaningful insights from textual data.

The practical significance of our LDA-GMM-CorEx model is extensive. The hybrid model could allow IS professionals to implement systems that can process and analyze data in real time. This is particularly useful in applications, such as social media monitoring, real-time market analysis, and dynamic content curation, in which data elements frequently change and require immediate attention. The ability to adapt dynamically to these changes enhances the utility of IS in providing timely and relevant insights for decision-making. Furthermore, the synergy between LDA, the GMM, and CorEx allows scalable analysis without significantly increasing computational resources, making it feasible to handle the large-scale datasets commonly encountered on digital platforms.

The use of CorEx to integrate domain-specific anchor words significantly improves the interpretability of the topics generated, providing actionable insights for strategic decision-making. In practice, this means that the insights derived from data analytics are clear and aligned with the organization's specific needs and goals. This feature enables managers and decision-makers to utilize these insights in strategic planning and operational adjustments, directly impacting management practices and business outcomes. In addition, the hybrid model's refined semantic analysis capabilities allow firms to extract more precise and contextually relevant insights from customer data, improving personalization of services and products.

Moreover, the model effectively manages lexical distribution and incorporates real-time domain expertise, addressing the challenges posed by digital turbulence—such as the rapid evolution of information and the constant shifting of digital content themes. It serves as a robust tool for analyzing large volumes of unstructured text data, enabling clearer insights and more actionable information extraction. This capability is critical for organizations and systems seeking stability and informed decision-making amidst the chaos of digital information overload.

5.2. Limitations and future research

This study has certain limitations. First, given that the analysis focuses on two online text datasets, it provides a certain reference value for applying the LDA-GMM-CorEx topic model in IS research. However, the generalization of the model could be extended to a larger research field. It may apply to any domain where strong interpretability in topic modeling or prior domain knowledge is required. Therefore, future studies should test its validity on additional datasets to verify its applicability. Second, although our model demonstrated significant improvement of nonstandard datasets, further exploration is necessary for corpora of other languages and those with mixed long and short texts. Finally, the combination of LDA, the GMM, and CorEx, increases computational complexity. For instance, LDA involves iterative computations over large document collections to infer topic distributions, and the GMM requires estimating multiple parameters, increasing with the number of components in the mixture. CorEx adds another layer of complexity by seeking maximal mutual information, which involves nonlinear optimization. To address these challenges, future research could implement incremental learning methods, allowing the model to

update with new data without reprocessing the entire dataset, which could significantly reduce computational demands. Additionally, we will explore the potential for simplifying the model by reducing the number of GMM parameters or components or by integrating more computationally efficient variants of LDA and CorEx.

CRediT authorship contribution statement

Yuting Jiang: Conceptualization, Data curation, Methodology, Formal analysis, Writing – original draft, Visualization, Writing – review & editing. **Mengyao Fu:** Supervision, Data curation, Project administration, Writing – review & editing. **Jie Fang:** Supervision, Visualization, Writing – review & editing. **Matti Rossi:** Supervision, Writing – review & editing. **Yuting Wang:** Supervision, Resources, Writing – review & editing. **Chee-Wee Tan:** Conceptualization, Supervision, Resources, Writing – review & editing, Project administration, Funding acquisition.

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